In-Country Working Session: FASTR Implementation & RMNCAH-N Service Monitoring Analysis

January 27-30, 2026 | Lusaka

GFF FASTR Team

FASTR
FREQUENT ASSESSMENTS & SYSTEMS TOOLS FOR RESILIENCE

Agenda

Day 1 -- Laying the Foundation: Introducing FASTR and Configuring the Analytics Platform

Time	Agenda	Facilitator/Presenter
Opening Session		
08:30-09:00	Participant registration	MoH team
09:00-09:10	Welcome and opening remarks	MoH team
09:10-09:20	Icebreakers/Introductions	MoH team
09:20-09:35	Overview of agenda, workshop objectives	GFF FASTR team
Session 1: Overview of the FASTR approach		
09:35-10:30	Overview: FASTR Approaches	GFF FASTR team
Session 2: HMIS data extraction		
10:30-11:30	Data extraction: Rationale and methods	GFF FASTR team
Session 3: Introduction to the FASTR analytics platform		
11:30-12:30	Introduction to the FASTR analytics platform	GFF FASTR team
12:30-14:00	*Lunch Break*	
14:00-14:30	Getting participants into the platform	GFF FASTR team
Session 4: Configuring the FASTR analytics platform		
14:30-16:30	Configuring the analysis platform	GFF FASTR team

Agenda

Day 2 -- Building the Analysis: Applying FASTR Methods and Generating Outputs

Time	Agenda	Facilitator/Presenter
09:00-09:15	Overview of Day 2 agenda	GFF FASTR team
Session 5: Overview of FASTR methods and analytical outputs		
09:15-10:15	Data quality, service utilization, coverage	GFF FASTR team
Session 6: Creating a project		
10:15-11:15	Project creation and settings	GFF FASTR team
Session 7: Creating visualizations		
11:15-12:30	Creating and editing visualizations	GFF FASTR team
12:30-14:00	*Lunch Break*	
Session 8: Creating reports		
14:30-16:30	Practice creating and editing reports	GFF FASTR team

Agenda

Day 3 -- From Analysis to Action: Interpreting Results and Using FASTR for Decision-Making

Time	Agenda	Facilitator/Presenter
09:00-09:15	Overview of Day 3 agenda	GFF FASTR team
Session 8: Interpretation of visualizations		
09:15-10:15	Approaches to support interpretation	GFF FASTR team
Session 9: Creating a Q4 2025 report		
10:15-12:30	Creating short and long reports with country context	GFF FASTR team
12:30-14:00	*Lunch Break*	
Session 9 (cont'd): Creating a Q4 2025 report		
14:00-15:00	Continue report creation with country context	GFF FASTR team
Session 10: Presenting reports		
14:30-15:30	Present reports, group feedback	GFF FASTR team

Agenda

Day 4 -- Designing the Health Facility Assessment

Time	Agenda	Facilitator/Presenter
09:00-09:15	Overview of Day 4 agenda	GFF FASTR team
Session 12: Overview of FASTR HFA phone survey		
09:15-10:15	HFA overview and questionnaire adaptation guidelines	GFF FASTR team
Session 13: Questionnaire adaptation to the Zambian context		
10:15-12:30	Review questionnaire + hands-on adaptation	GFF FASTR team
12:30-14:00	*Lunch Break*	
Session 14: Questionnaire adaptation (cont'd)		
14:00-15:00	Continue questionnaire adaptation (in groups)	GFF FASTR team
Session 15: Discussion: HFA adapted questionnaire		
14:30-15:30	Discuss adapted questionnaire in plenary	GFF FASTR team
Session 16: Discussion: HFA priorities and data use		
15:30-16:30	HFA priorities and data use case in Zambia	GFF FASTR team
Session 17: Action planning and wrap-up		
16:30-17:00	Key messages and wrap-up	GFF FASTR team

Workshop Objectives

- Prepare MoH for participation in the multi-country workshop
- Strengthen capacity for disruption analysis and data extraction
- Configure the FASTR analytics platform for Zambia
- Produce the first quarterly report
- Adapt the phone survey questionnaire to the Zambian context
- Use data downloader tools for DHIS2

Scope of Work

Disruption Analysis Activities

- Data extraction and configuration of analytics platform
- Produce first quarterly report
- Review results and contextualize findings (with relevant program teams)

Phone Survey Activities

- Adaptation of questionnaire (with program input)

Capacity Building

- Training on data downloader for all with DHIS2 access
- Hybrid format: face-to-face and online

Expected Outputs

- Adapted phone survey questionnaire
- First quarterly report draft
- Trained MoH team on data downloader tools
- Roadmap for participation in Abuja workshop

Session 1: Overview of the FASTR approach

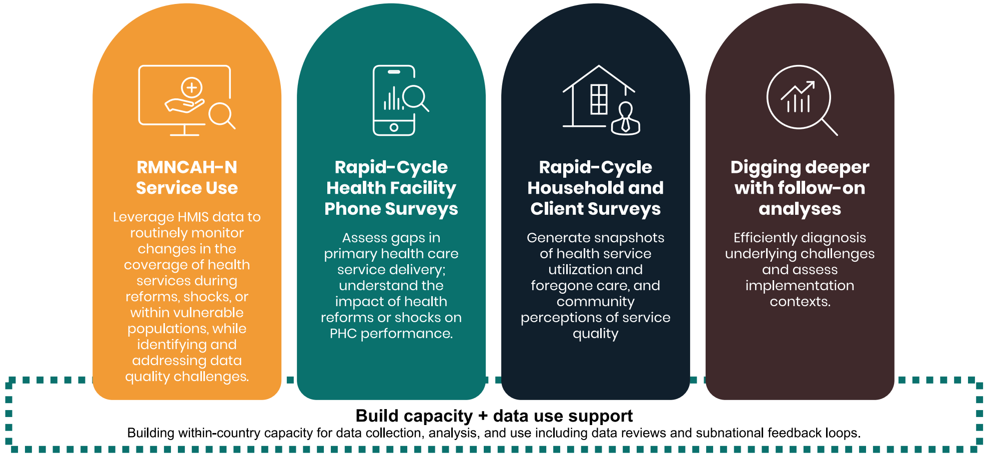
Introduction to FASTR

The Global Financing Facility (GFF) supports country-led efforts to strengthen the use of timely data for decision-making, with the goal of improving primary healthcare (PHC) performance and RMNCAH-N outcomes.

Frequent Assessments and Health System Tools for Resilience (FASTR) is the GFF's rapid-cycle analytics framework for monitoring health system performance using high-frequency data.

FASTR brings together four complementary technical approaches:

1. Routine HMIS data analysis
2. Health facility phone surveys
3. High-frequency household phone surveys
4. Follow-on, problem-driven analyses



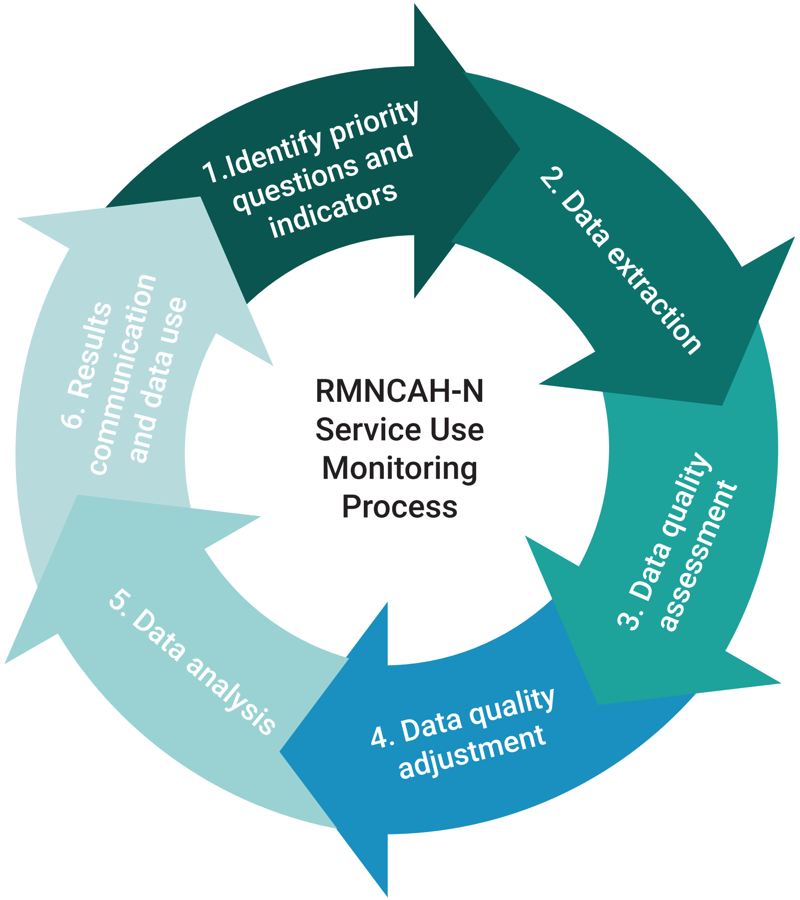
RMNCAH-N service use monitoring

Rapid-cycle approaches using routine health management information system (HMIS) data can be used to:

- **Evaluate the quality of HMIS data** both at the national and sub-national levels to help address gaps in quality and completeness
- **Measure monthly changes** in the utilization of critical health services, enabling a rapid response to challenges and the opportunity to gain insights on the progress of reforms
- **Compare service coverage trends** with country targets, facilitating the continuous monitoring of RMNCAH-N progress

How does it work?

Through collaboration with Ministries of Health, the GFF assists countries in developing and reviewing regular analyses of HMIS data, focusing on specific priority health service indicators linked to national health care reforms, as well as GFF and World Bank investments.



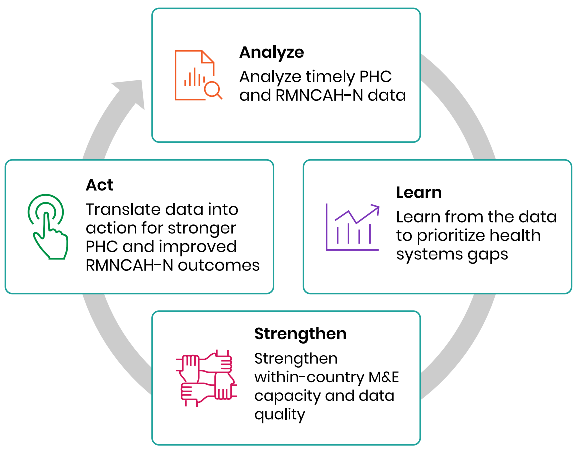
Why rapid-cycle analytics?

Routine health information systems are a critical source of data, but they are often underused due to concerns about data quality and long delays between data collection and analysis. Traditional household and facility surveys, while essential, are resource-intensive and infrequent.

FASTR's rapid-cycle analytics address this gap by providing:

- Timely insights aligned with country decision cycles
- Continuous learning rather than one-off assessments
- Direct feedback loops between data, analysis, and action

FASTR's rapid-cycle analytics approach: Analyze, learn, strengthen, act



Focus of the analysis

Core indicators

FASTR prioritizes a core set of RMNCAH-N indicators that:

- Represent key service delivery contacts across the continuum of care
- Have relatively high reporting completeness and volumes
- Serve as proxies for broader service delivery performance

Outpatient consultations are included as a proxy for overall health service use. The indicator set can be expanded to reflect country-specific priorities.

Core data quality metrics

Analysis is anchored in a standardized set of data quality metrics, including:

- Reporting completeness
- Extreme value (outlier) detection
- Consistency across related indicators

These metrics are summarized into an overall data quality score to support interpretation and comparison across areas.

Session 2: HMIS data extraction

Why extract data from DHIS2?

Data quality adjustment

The FASTR approach focuses on data quality adjustments to expand the analyses countries can do with DHIS2 data and to generate more robust estimates.

The FASTR methodology includes specific approaches to:

- Identify and adjust for outliers
- Adjust for incomplete reporting
- Apply consistent data quality metrics

These adjustments require processing that cannot be done within DHIS2's native analytics.

Why extract data from DHIS2?

Analysis complexity

The FASTR approach uses more advanced statistical methods, such as regression analysis, which are not available in DHIS2. While DHIS2 can plot trends over time using raw data, FASTR can go further by:

- Identifying significant increases or decreases in service volume
- Adjusting for data quality issues
- Accounting for expected seasonal variations
- Comparing key periods, such as before and after a reform

The choice between DHIS2 and the FASTR approach should be guided by the specific purpose of your analysis.

Data format and granularity

Data should be downloaded for each **indicator of interest**, at **facility level**, and **monthly** for the **period of interest**.

- Data should be saved in **long format** meaning each row represents a single observation or measurement
- Data should be saved in **.csv format** and can be saved in either a single .csv file or multiple .csv files

Why monthly facility level data?

We want to use the most granular data we have access to in order to make more fine tuned assessments for data quality. Using monthly facility level data allows us to conduct the most robust analysis.

Key variables

The data extracted should include the following required elements:

Element	Description
Org units	Organizational unit identifier
Period	Time period of the data
Indicator name	Name of the indicator
Total/count	The aggregated value

How much data?

Initial FASTR analysis

- Download approximately **five years** of historical data
- Exact period depends on data availability and consistency in indicator definitions

Routine update to FASTR analysis

- Download new data covering the most recent months not previously included (usually **three months** for quarterly implementation)
- Include the **three preceding months** as recent data is often subject to changes due to late reporting or data quality adjustments

Data extraction tools

We offer two tools for bulk DHIS2 data extraction:

API Script (Google Colab)

- Input login credentials, specify timeframes, indicators, and administrative levels
- Download data as a .csv file

Data Downloader

- More intuitive, streamlined interface
- Recommended for most users

Both tools enable efficient data extraction, and we provide training resources to support their use.

DHIS2 Data Downloader

The Data Downloader is a desktop application for extracting data from DHIS2.

Key features:

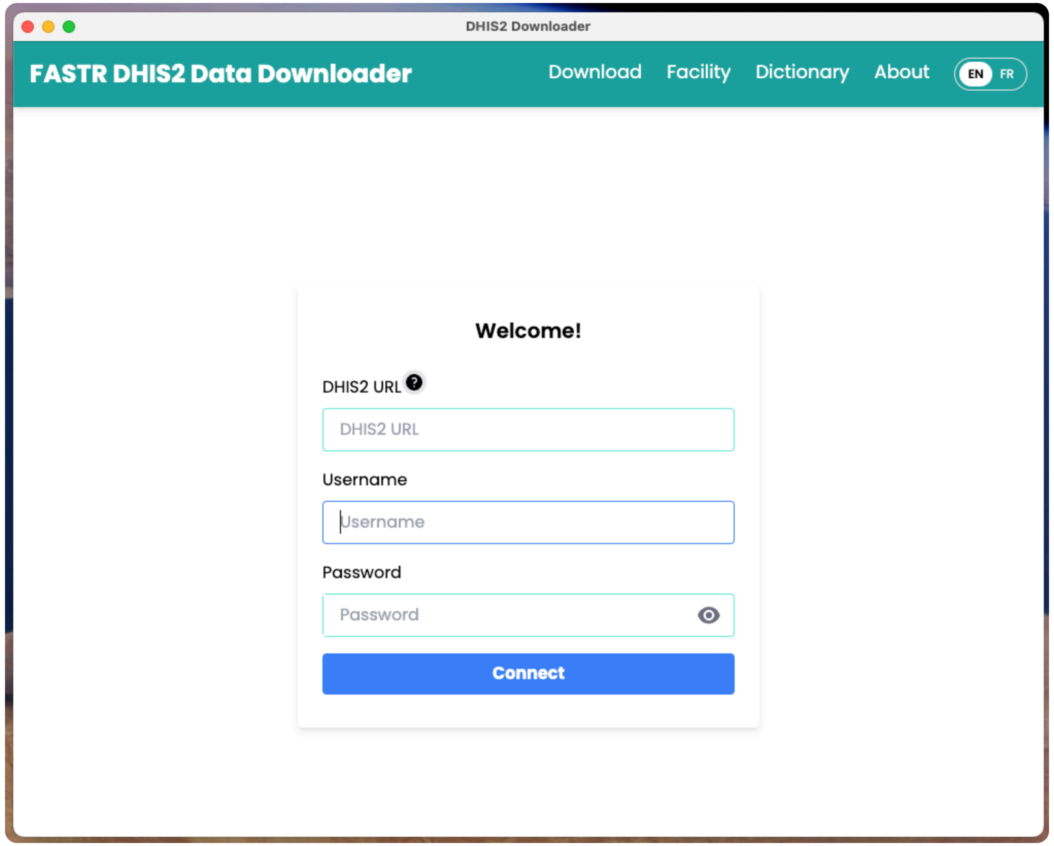
- Connect to any DHIS2 instance
- Browse and select data elements and indicators
- Download facility-level data in CSV format
- Maintain download history

Download from GitHub:

https://github.com/worldbank/DHIS2-Downloader/releases/

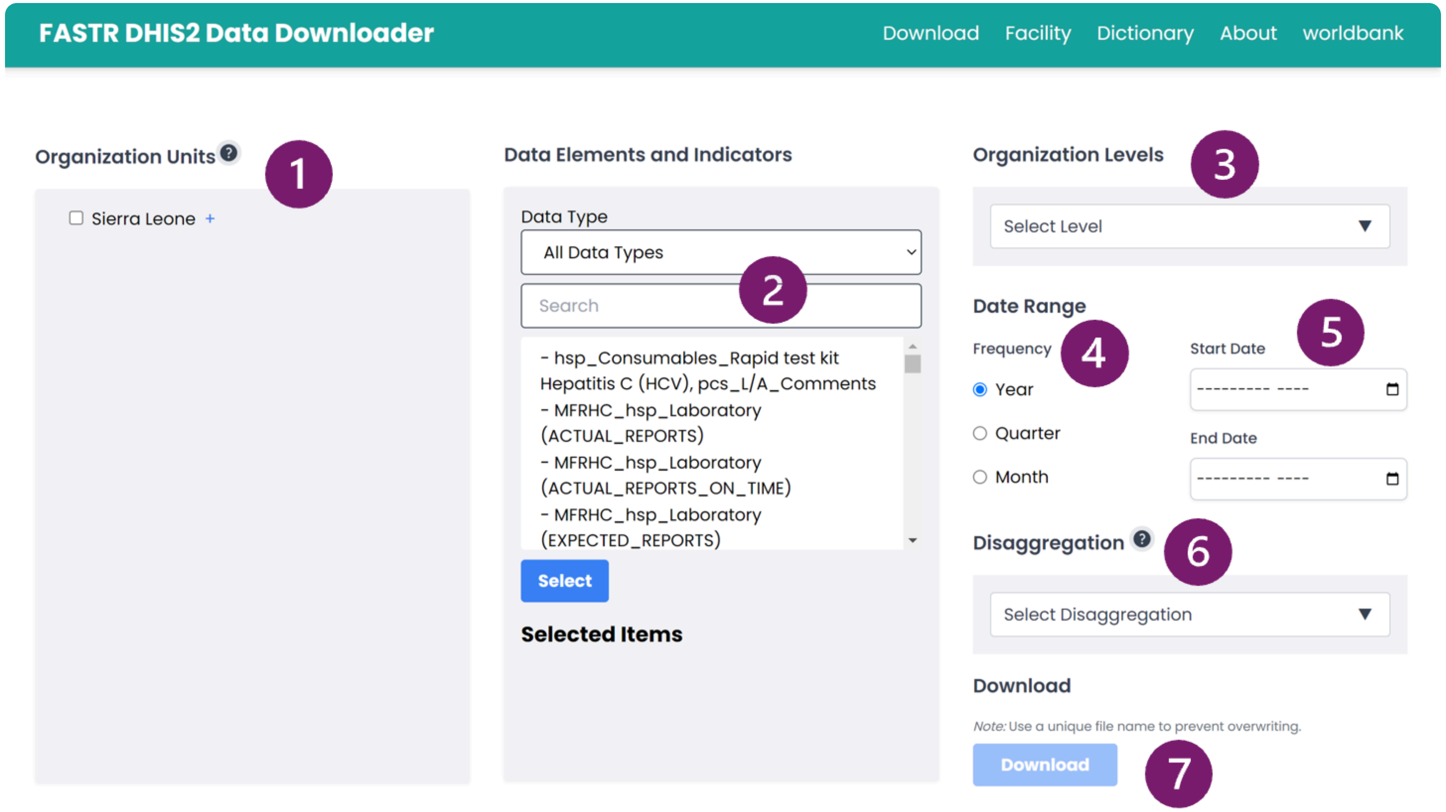
Facilitator will demonstrate the Data Downloader

Data Downloader: Login



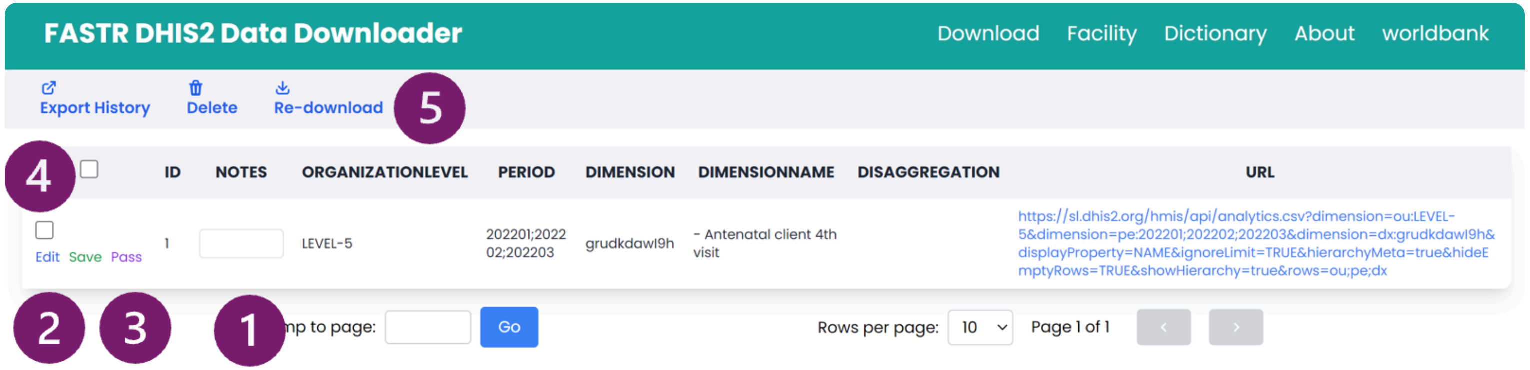
Enter your DHIS2 instance URL and credentials to connect.

Data Downloader: Overview



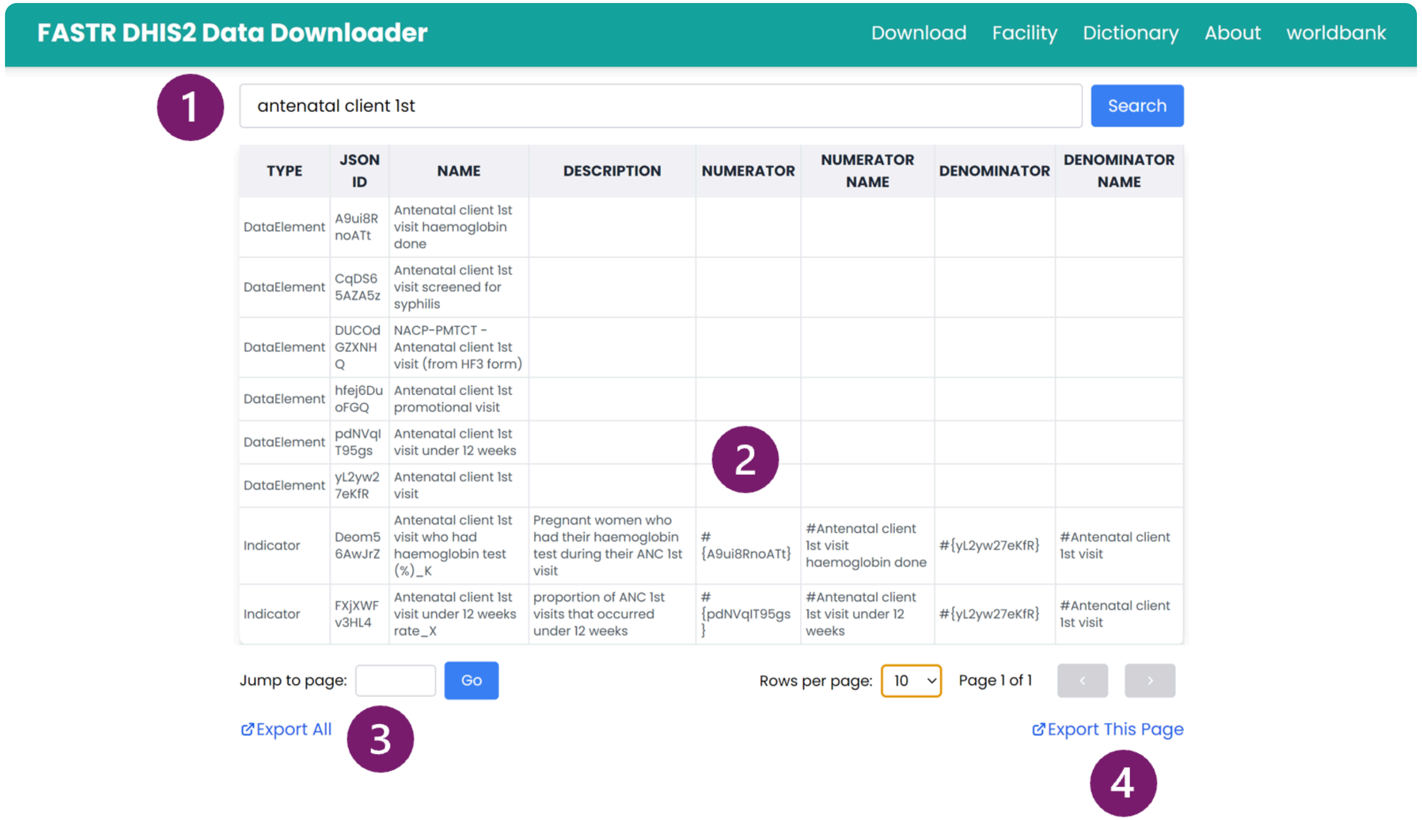
The main interface displays available data elements and download options.

Data Downloader: Download history



Track previous downloads and re-download data as needed.

Data Downloader: Data dictionary



TYPE	JSON ID	NAME	DESCRIPTION	NUMERATOR	NUMERATOR NAME	DENOMINATOR	DENOMINATOR NAME
DataElement	A9ui8RnoATt	Antenatal client 1st visit haemoglobin done					
DataElement	CqDS65AZA5z	Antenatal client 1st visit screened for syphilis					
DataElement	DUCOdGZXNHQ	NACP-PMTCT - Antenatal client 1st visit (from HF3 form)					
DataElement	hfej6DuoFGQ	Antenatal client 1st promotional visit					
DataElement	pdNVqlT95gs	Antenatal client 1st visit under 12 weeks					
DataElement	yL2yw27eKfR	Antenatal client 1st visit					
Indicator	Deom56AwJrZ	Antenatal client 1st visit who had haemoglobin test (%)_K	Pregnant women who had their haemoglobin test during their ANC 1st visit	#{A9ui8RnoATt}	#Antenatal client 1st visit haemoglobin done	#{yL2yw27eKfR}	#Antenatal client 1st visit
Indicator	FXjXWFv3HL4	Antenatal client 1st visit under 12 weeks rate_X	proportion of ANC 1st visits that occurred under 12 weeks	#{pdNVqlT95gs}	#Antenatal client 1st visit under 12 weeks	#{yL2yw27eKfR}	#Antenatal client 1st visit

Browse and search available data elements and indicators from your DHIS2 instance.

Data Downloader: Facility list

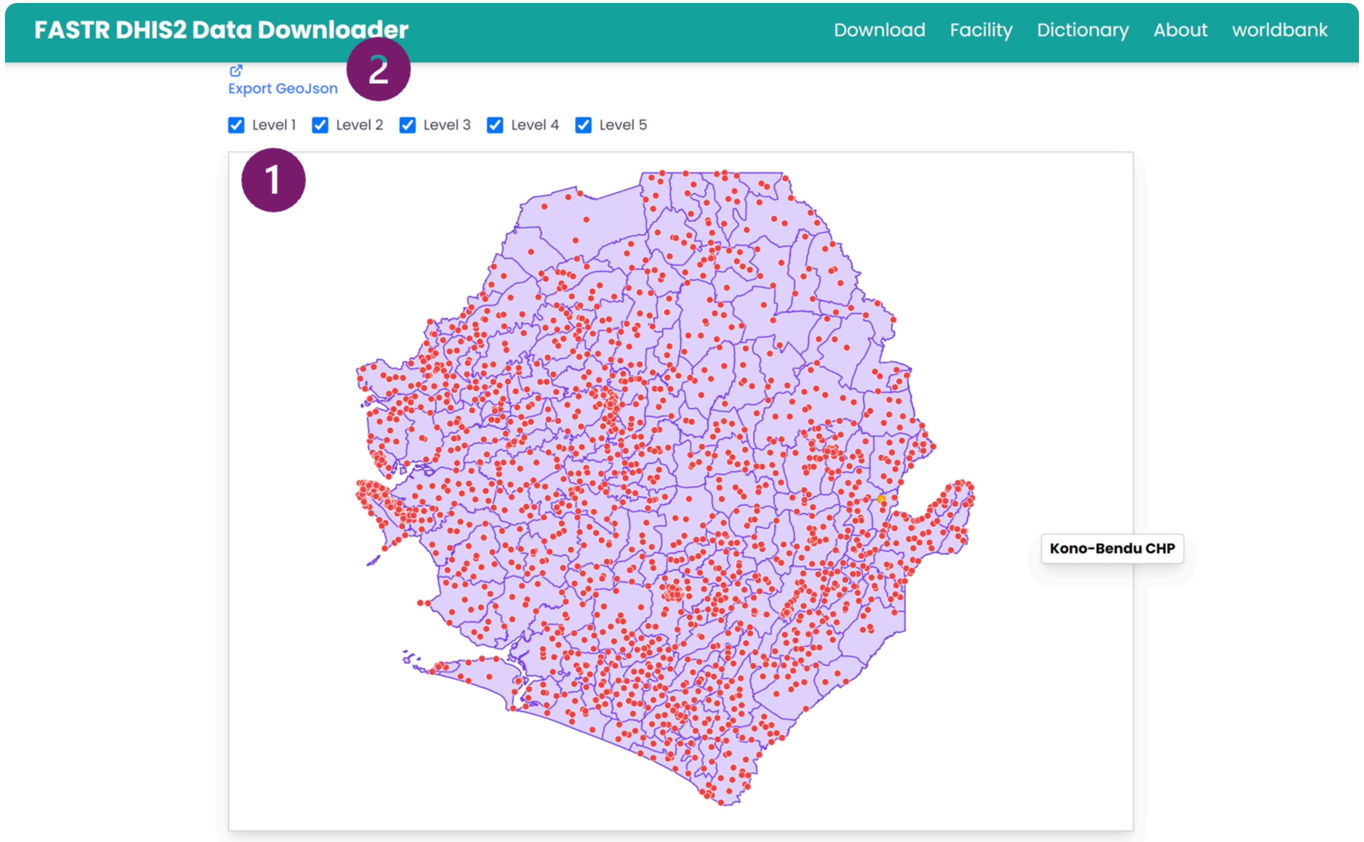
FASTR DHIS2 Data Downloader Download Facility Dictionary About worldbank

Export CSV

LEVEL	ORGUNITLEVEL 1	ORGUNITLEVEL 2	ORGUNITLEVEL 3	ORGUNITLEVEL 4	NAME	ID	COORDINATES	ORGANIZATION UNIT GROUPS
5	Sierra Leone	Kono District	Kono District Council	Kamara Chiefdom	Kamara Communities	A0UqOtjW8Of	-11.026827, 8.721628	Communities (Placeholders), Public
5	Sierra Leone	Kailahun District	Kailahun District Council	Luawa Chiefdom	Bailu Port of Entry	A10jHz59FN2	No geometry available	Port of Entry, Public
5	Sierra Leone	Pujehun District	Pujehun District Council	Galliness Chiefdom	Kpowubu MCHP	A1OP0onYJiX	-11.5709, 7.342915	MCHP, PMTCT Facilities, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Koinadugu District	Koinadugu District Council	Kamukeh Chiefdom	Thellia Port of Entry	A1pv9kyBwU4	-11.7989, 9.9986	Port of Entry, Public
5	Sierra Leone	Falaba District	Falaba District Council	Kamadugu Yiraia Chiefdom	Dankawalie CHC	A1zGIPUnnJp	-11.331542, 9.629802	CHC, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Pujehun District	Pujehun District Council	Peje Chiefdom	Pejewa (Futa Peje) MCHP	A6dVJNJNJLU	-11.506295, 7.587257	MCHP, PMTCT Facilities, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Kailahun District	Kailahun District Council	Kissi Kama Chiefdom	Sangha Port of Entry	A81tty0wW9c	-10.415372, 8.436378	Port of Entry, Public
5	Sierra Leone	Kenema District	Kenema City Council	Kenema City	Lango Town MCHP	A82aB5gRfth	-11.17263, 7.860986	MCHP, PMTCT Facilities, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Bonthe District	Bonthe District Council	Sittia Chiefdom	Mbokie MCHP	A9MVWMOILZh	-12.6161, 7.6369	MCHP, PHUs, eIDSR-Reporting facilities, Public
5	Sierra Leone	Western Area Urban District	Freetown City Council	East 3 Zone	Egyptian (Calaba Town) Clinic	A9zxwZDuOHp	-13.17487, 8.46455	eIDSR-Reporting facilities, Clinic, Private

View and filter facilities by administrative level and organization unit.

Data Downloader: Facility map



Visualize facility locations on an interactive map.

Session 3: Introduction to the FASTR analytics platform

Introduction to the FASTR Analytics Platform

The FASTR analytics platform is a web-based tool for data quality assessment, adjustment, and analysis of routine health data.

Key features:

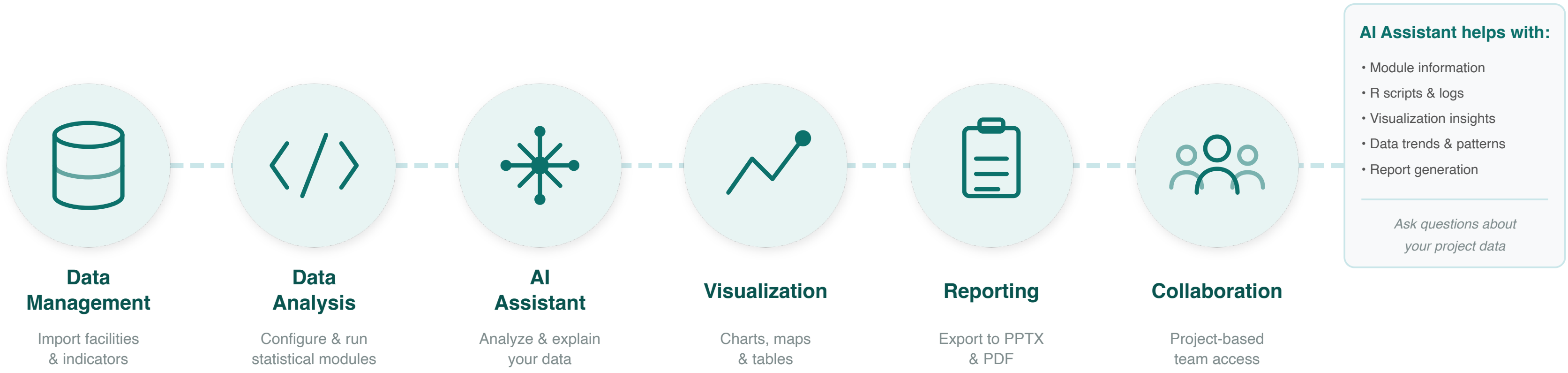
- Upload and analyze data from DHIS2 and other sources
- Built-in statistical methods for data quality adjustment
- User-friendly interface for running analyses
- Flexible visualization and export options

In this session, we will provide a conceptual walkthrough of the platform and its capabilities.

Platform Capabilities

FASTR Analytics Platform

Key Capabilities



Data flows from import through analysis to shareable outputs

Data flows from import through analysis to shareable outputs.

Live Demo: Platform Access & Roles

In this demo, we will:

- Navigate to the FASTR platform
- Explore user roles: Administrator, Editor, Viewer
- Review user management and permissions
- Understand the workflow for uploading data and making analytical decisions

Facilitator will demonstrate in the live platform

🍽️ Lunch Break

90 minutes

Back at 14:00

Session 4: Configuring the FASTR analytics platform

Activity: Setting Up Admin Areas

In this hands-on session, we will configure:

- Admin areas (regions, districts)
- Facility structure
- Indicator definitions

Participants will work directly in the platform

Activity: Importing Data

In this hands-on session, we will:

- Review data format requirements
- Walk through the import process
- Handle validation and error checking

Participants will import their country's data

Activity: Installing and Running Modules

In this hands-on session, we will:

- Review available analysis modules
- Install required modules
- Run initial analyses

Participants will configure and run modules on their data

Day 2

Recap

On Day 1, we covered:

- The FASTR approach and rapid-cycle analytics methodology
- Data extraction from DHIS2 using the Data Downloader
- Introduction to the FASTR Analytics Platform
- Platform configuration: admin areas, data import, modules

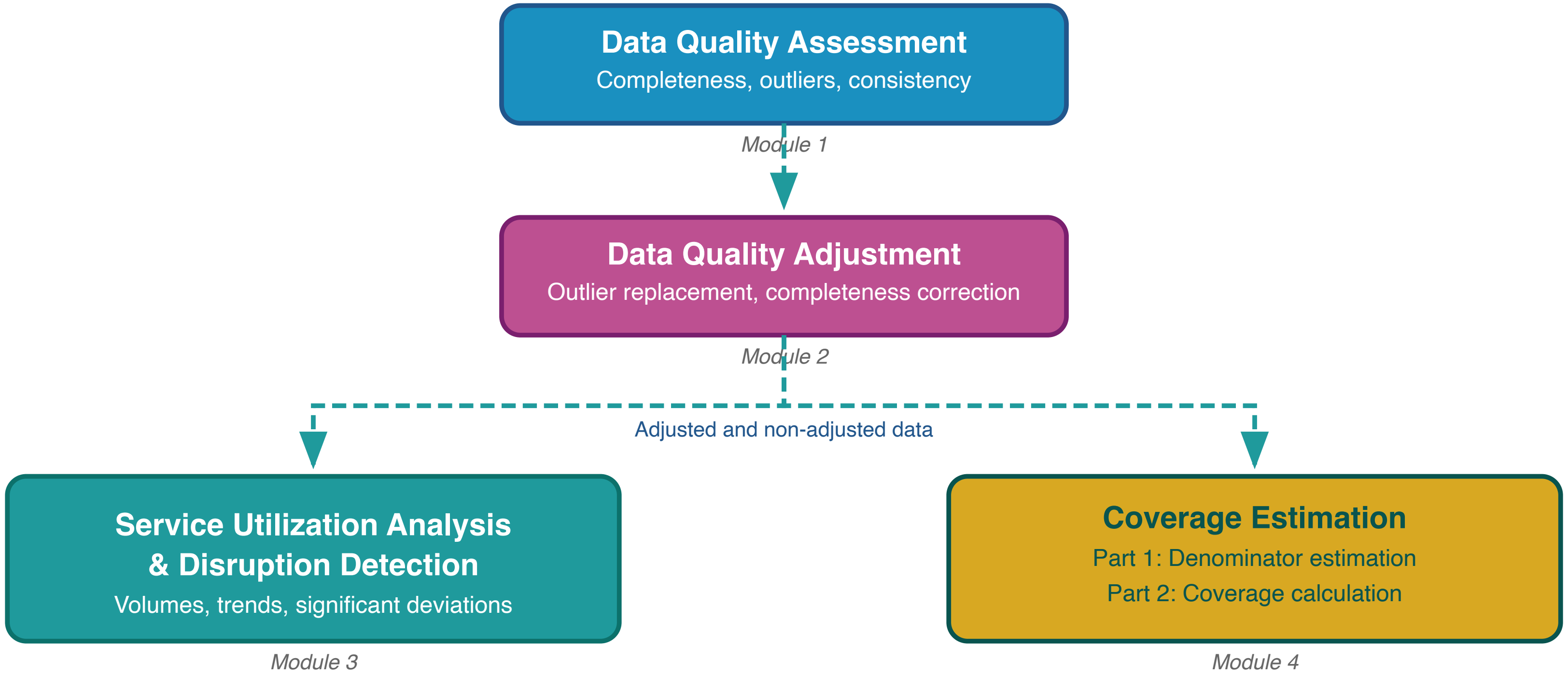
Day 2 Focus

Today we will:

- Explore FASTR methods for data quality and analysis
- Create and configure projects
- Build visualizations
- Generate reports

Session 5: Overview of FASTR methods and analytical outputs

FASTR analytical pipeline



The FASTR analysis follows a sequential workflow where each step builds on the previous:

1. **Assess data quality** - Identify issues with completeness, outliers, and consistency
2. **Adjust for quality issues** - Apply corrections to improve data reliability
3. **Analyze adjusted data** - Generate service utilization and coverage estimates

Data quality assessment - Module 1

Evaluating the reliability of routine health information system data

Rationale for data quality assessment

Challenge: Routine health facility data may contain quality limitations:

- Reported values may fall outside plausible ranges
- Reporting gaps affect data completeness
- Inconsistencies exist between related indicators

Implications: Data quality limitations affect decision-making

- Inaccurate assessments of service delivery trends
- Misidentification of areas requiring intervention
- Suboptimal resource allocation

Objectives of data quality assessment

Objective 1: Enable analytical adjustment

Systematic data quality assessment supports the application of targeted adjustments, enhancing the utility of HMIS data for evidence-based decision-making.

Objective 2: Monitor data quality trends

Data quality assessment enables ongoing monitoring to:

- Inform indicator selection based on quality profiles across the HMIS
- Guide targeted data quality interventions and supportive supervision in areas with weaker data quality
- Evaluate the effectiveness of data quality improvement initiatives over time

Core dimensions of data quality

1. Completeness

Are health facilities submitting reports consistently?

2. Outlier prevalence

Are reported values within plausible ranges?

3. Internal consistency

Do related indicators demonstrate expected relationships?

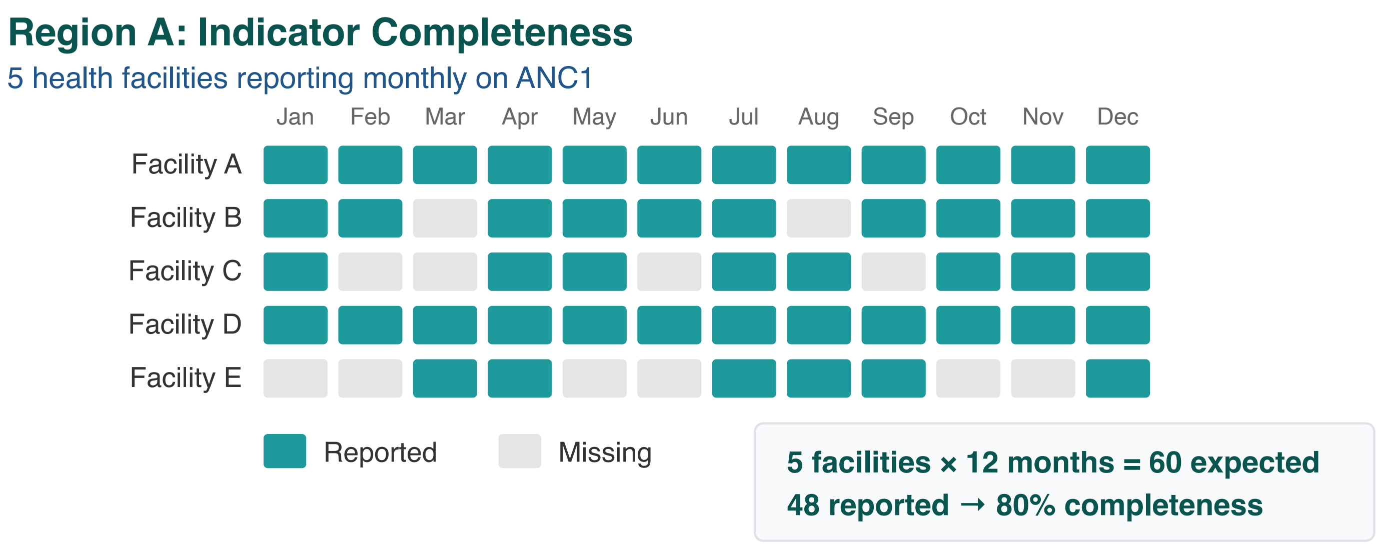
These three dimensions provide a comprehensive assessment of data reliability for analytical purposes.

Indicator completeness

Indicator completeness measures the extent to which facilities that are supposed to report data on the selected core indicators are in fact doing so.

Higher completeness improves reliability of the data, especially when completeness is stable over time.

This is different from overall reporting completeness in that it looks at completeness of specific data elements and not only at the receipt of the monthly reporting form.



Definition of indicator completeness

For the FASTR analysis, completeness is defined as:

The percentage of reporting facilities each month out of the total number of facilities expected to report.

- A facility is deemed to be "reporting" if there is a non-missing, non-zero value recorded for the indicator and month
- A facility is expected to report if it has reported any volume for that indicator anytime within a year

Notes on completeness

- A high level of completeness does not necessarily indicate that the HMIS is representative of all service delivery in the country as some services may not be delivered in facilities, or some facilities may not report
- For countries where the DHIS2 system does not store 0's, indicator completeness may be underestimated if there are many low-volume facilities for a given indicator
- Facilities that do not report for six or more consecutive months at the beginning or end of their reporting period are classified as **inactive** rather than incomplete. This prevents penalizing facilities that have not yet begun reporting or have permanently ceased operations

For a given indicator in a given time period, the percent of monthly values that are complete:

% complete = # monthly values that are complete / total N of monthly values

Indicator Completeness

Percentage of facility-months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

- 90% or above
- 80% to 89%
- Below 80%

Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities

Outliers

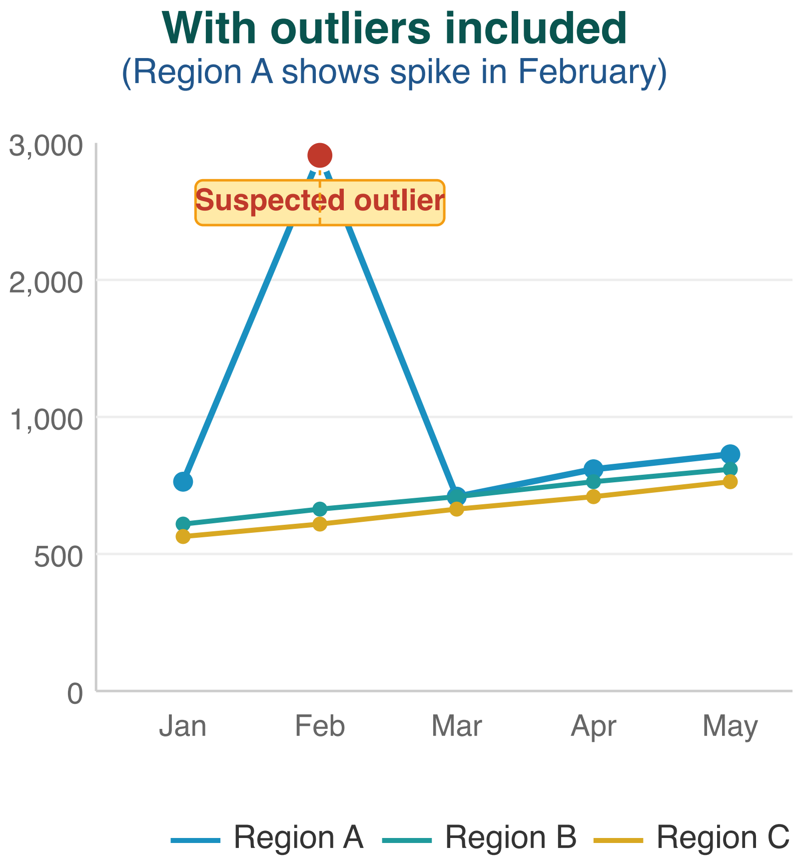
The presence of outliers examines whether a data point in a series of values is extreme (either abnormally high or low) in relation to others in the series.

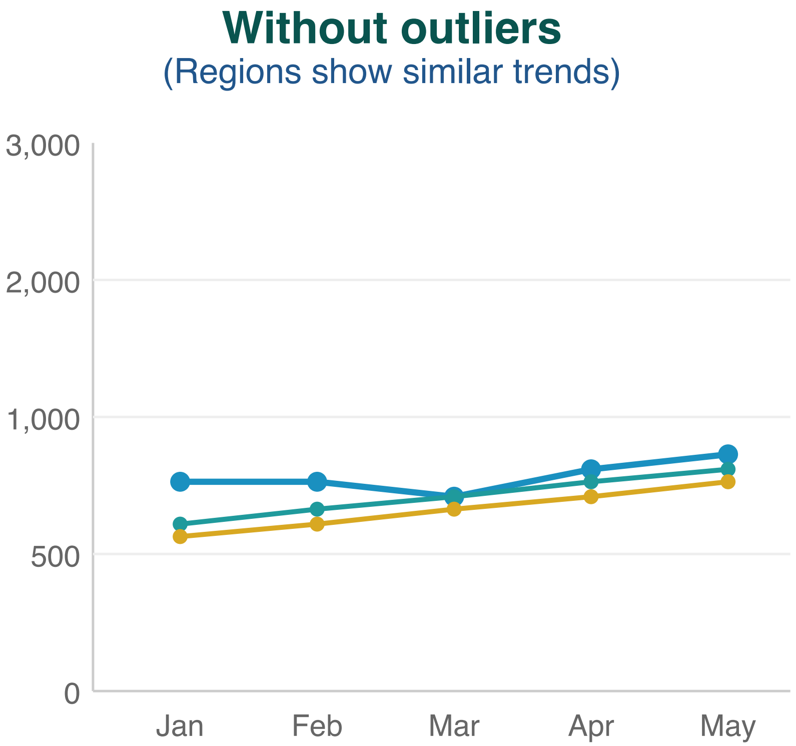
Outliers can be the result of changes in programmatic activities (such as an intensified campaign) or can be data quality problems.

For the FASTR analysis, we identify outliers which are suspiciously high values compared to the usual volume of services reported by the facility (e.g., low values are not identified as outliers in the FASTR analysis).

Outlier illustration

Region A displays an anomalous spike in February that substantially exceeds values reported by other regions — indicative of a data entry error or reporting issue.





Outliers can distort the true picture of service trends

Outlier detection methodology

Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator.

An outlier is defined as:

A value greater than **10 times the median absolute deviation (MAD)** from the monthly median value for the indicator in each time period, **OR** a value for which the proportional contribution in volume for a facility, indicator, and time period is **greater than 80%**

AND for which:

- The volume is **greater than or equal to the median**
- The volume is **not missing**
- The volume is **greater than 100**

Outliers: Percent of monthly values that are outliers

For a given indicator in a given time period, the percent of monthly values that are outliers:

% outliers = # monthly values that are outliers / total N of monthly values

Outliers

Percentage of facility-months that are outliers, May 2024 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
Region 001	0.6%	0.2%	0.2%	1.9%	1.2%	0.6%	0.9%	0.6%	0.9%
Region 002	0.6%	0.0%	1.2%	1.0%	1.8%	0.2%	0.2%	0.1%	2.7%
Region 003	0.5%	0.4%	0.1%	0.0%	0.1%	0.1%	0.4%	0.6%	1.4%
Region 004	0.4%	0.0%	0.0%	0.8%	0.0%	0.5%	0.8%	0.9%	0.1%
Region 005	0.5%	0.9%	0.5%	0.8%	0.5%	0.6%	0.8%	0.3%	3.3%
Region 006	0.4%	0.0%	0.1%	0.3%	0.1%	1.3%	2.3%	2.5%	0.6%

3% or above
1% to 2%
Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100.

Consistency between related indicators

Program indicators with a predictable relationship are examined to determine whether the expected relationship exists between them. In other words, this process examines whether the observed relationship between the indicators, as shown in the reported data, is that which is expected.

Indicator pairs assessed

FASTR assesses the following pairs of indicators to measure internal consistency:

Indicator pair	Expected relationship
ANC1 / ANC4	Ratio should be ≥ 0.95
Penta1 / Penta3	Ratio should be ≥ 0.95
BCG / Facility delivery	Ratio should be within 30% (i.e. ≥0.7 and ≤1.3)

These pairs of indicators have expected relationships.

We expect the number of pregnant women receiving a first ANC visit will always be higher than the number of pregnant women receiving a fourth ANC visit.

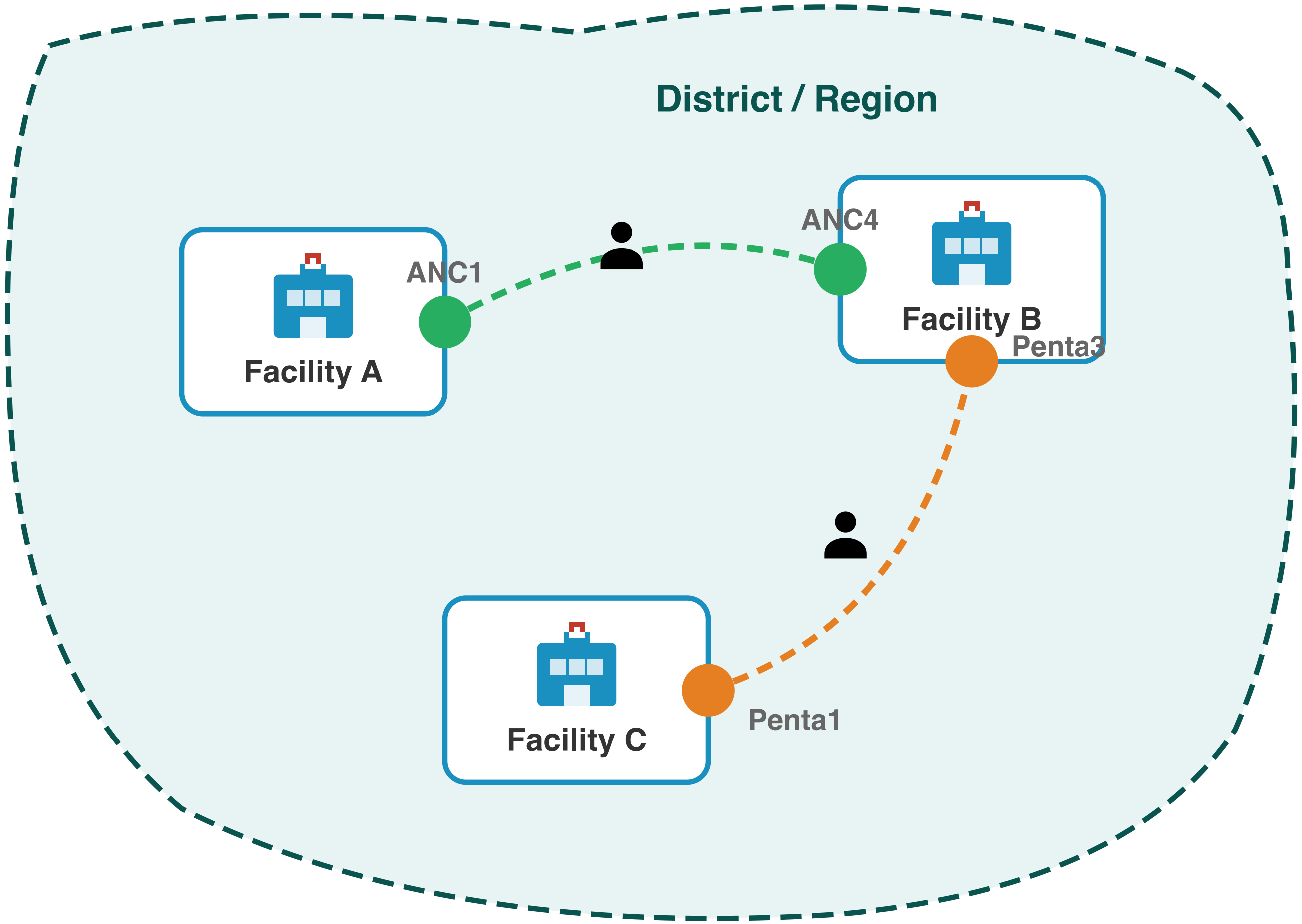
BCG is a birth dose vaccine so we expect that these indicators will be equal. However, we recognize there may be more variability in this predicted relationship thus we set a range of within 30%.

Why assess consistency at district level?

Patients often access different services at different facilities within a district:

- A woman may attend **ANC1** at a nearby health post, but travel to a health centre for **ANC4**
- A child may receive **Penta1** at a local clinic, but complete **Penta3** at a district hospital

Checking consistency at the facility level would miss these patterns. Aggregating to district level captures the complete picture of service utilization within a geographic area.



Internal consistency: FASTR output

Internal consistency

Percentage of sub-national areas meeting consistency benchmarks, May 2024 to Apr 2025

	ANC1 is larger than ANC4	Delivery is approximately equal to BCG	Penta1 is larger than Penta3
Region 001	100.0%	0.0%	97.2%
Region 002	100.0%	0.0%	100.0%
Region 003	100.0%	45.8%	83.3%
Region 004	100.0%	37.5%	87.5%
Region 005	100.0%	0.0%	87.5%
Region 006	100.0%	49.0%	84.4%

- 90% or above
- 80% to 89%
- Below 80%

Internal consistency assesses the plausibility of reported data based on related indicators. Consistency metrics are approximate - depending on timing and seasonality, indicator definitions, and the nature of service delivery and reporting, values may be expected to sit outside plausible ranges. Indicators which are similar are expected to have roughy the same volume over the year (within a 30% margin). The data in this analysis is adjusted for outliers.

Data quality summary score

A composite measure of data quality provides an overall view of how well a dataset meets quality standards.

By integrating multiple dimensions of data quality into a single score, it simplifies the interpretation of detailed information from several measures. This allows health systems to quickly assess the reliability of data, making it easier to identify trends and issues at a glance.

Definition of adequate data quality

For the FASTR analysis, we defined adequate data quality as:

- No missing indicator data for OPD, Penta1, and ANC1, where available, **AND**
- No outliers for OPD, Penta1, and ANC1, where available, **AND**
- Consistent reporting between Penta1/Penta3 and ANC1/ANC4

Overall DQA score: Percent of monthly values meeting all criteria

For a given indicator in a given time period, the percent of monthly values meeting all DQA criteria:

% adequate quality = # monthly values meeting all criteria / total N of monthly values

Overall DQA score

Percentage of facility-months with adequate data quality over time

	2022	2023	2024	2025
Region 001	60.8%	76.6%	76.4%	84.4%
Region 002	55.3%	61.2%	58.2%	52.0%
Region 003	69.3%	73.4%	60.6%	47.0%
Region 004	54.6%	70.7%	70.0%	84.9%
Region 005	57.9%	69.5%	56.6%	55.4%
Region 006	74.0%	84.7%	72.2%	71.7%

80% or above
70% to 79%
Below 70%

Adequate data quality is defined as: 1) No missing data or outliers for OPD, Penta1, and ANC1, where available 2) Consistent reporting between Penta1/Penta3 and ANC1/ANC4.

Mean DQA score: How close are we to adequate quality?

The mean DQA score shows how close a facility's data is to meeting all quality criteria. A score of **100% means the data passes** all DQA checks—no missing values, no outliers, and consistent reporting.

Mean DQA = (completeness & outlier score + consistency score) / 2

Lower scores indicate more data quality issues that need attention before the data can be used with confidence.

Mean DQA score

Average data quality score across facility-months

	2022	2023	2024	2025
Region 001	90.2%	95.5%	95.5%	96.5%
Region 002	87.3%	89.0%	88.2%	85.6%
Region 003	92.8%	94.8%	91.0%	84.3%
Region 004	88.0%	93.9%	93.0%	97.0%
Region 005	89.5%	93.6%	91.1%	88.6%
Region 006	93.2%	96.3%	93.4%	93.4%

80% or above
70% to 79%
Below 70%

Items included in the DQA score include: No missing data for 1) OPD, 2) Penta1, and 3) ANC1, where available; No outliers for 4) OPD, 5) Penta1, and 6) ANC1, where available; Consistent reporting between 7) Penta1/Penta3, 8) ANC1/ANC4, 9)BCG/Delivery, where available.

DQA module: Configuration parameters

Parameter	Description
Proportion threshold for outlier detection	Adjusts the threshold for proportional contribution to flag a facility-month as an outlier
Minimum count threshold for consideration	Defines the minimum count required for a facility-month to be considered an outlier
Number of MADs	Outliers are defined as observations which are greater than X times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period
Indicators subjected to DQA	Defines which indicators are included for assessment of outliers and completeness for inclusion in the DQA score
Consistency pairs used	Defines which indicator pairs are used for consistency analysis and the expected ratio ranges

Data quality adjustment - Module 2

Correcting outliers and imputing missing values to improve data reliability

Rationale for data quality adjustment

Routine HMIS data contain two common limitations that can distort analytical results:

Issue	Impact on analysis
Outliers	Extreme values create artificial spikes in service volumes
Incomplete reporting	Missing data creates artificial declines that do not reflect actual service delivery

FASTR addresses these limitations by replacing problematic values with estimates derived from each facility's historical reporting patterns.

Adjustment scenarios

To support transparency and sensitivity analysis, FASTR produces four parallel datasets:

Scenario	Description
Unadjusted	Original reported values
Outliers adjusted	Extreme values replaced
Completeness adjusted	Missing values imputed
Both adjusted	All corrections applied

Indicators excluded from adjustment

Certain indicators are excluded from the adjustment process:

- **Mortality indicators** (maternal deaths, neonatal deaths, under-5 deaths): These represent discrete events where smoothing or imputation is not appropriate
- **Low-volume indicators**: Indicators that never exceed 100 reported events in any month are excluded from adjustment

Outlier adjustment methodology

Outlier values are replaced using facility-specific historical data. The adjustment follows a hierarchical approach:

Priority	Method	Application
1	Centered 6-month average	3 months before + 3 months after the outlier
2	Forward 6-month average	When insufficient preceding data (e.g., start of series)
3	Backward 6-month average	When insufficient following data (e.g., end of series)
4	Same month, previous year	When rolling averages unavailable; useful for seasonal indicators
5	Facility historical mean	Mean of all valid values for this indicator at this facility

Deviance Due to Outliers

Percent change in volume due to outlier adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	0.0%	0.2%	4.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.2%
Bombali District	1.3%	0.0%	0.0%	2.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%
Bonthe District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.5%	0.0%	0.0%	0.0%
Falaba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.7%	0.0%	0.0%	2.2%	0.0%	0.0%
Kailahun District	0.0%	0.0%	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	4.5%	2.1%	0.2%	0.2%
Kambia District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Karene District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.7%	1.3%	5.7%	4.7%	0.0%	0.0%
Kenema District	0.0%	0.0%	0.0%	0.0%	0.0%	1.2%	1.7%	1.4%	1.8%	0.4%	0.0%	0.0%
Koinadugu District	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	0.0%	2.6%	1.9%	0.0%	0.0%
Kono District	0.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.2%	0.6%	2.0%	1.8%
Moyamba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.7%	0.4%	0.1%	0.1%
Port Loko District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Pujehun District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.3%
Tonkolili District	0.0%	0.3%	0.0%	0.0%	0.0%	0.6%	0.0%	0.0%	1.5%	1.4%	0.1%	0.1%
Western Area Rural District	0.0%	0.0%	0.7%	1.5%	1.4%	0.5%	1.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Western Area Urban District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.6%	0.9%	1.1%	0.5%	1.5%	0.3%

3% or above
1% to 2%
Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100. The deviance is the difference in volume after removing the outlier. High levels of deviance can affect the plausiability of the data.

Completeness adjustment methodology

For months identified as incomplete or missing, values are imputed using the same 6-month rolling average approach applied to outlier adjustment.

Priority	Method	Application
1	Centered 6-month average	When sufficient data exists before and after the gap
2	Forward 6-month average	For gaps at the start of the time series
3	Backward 6-month average	For gaps at the end of the time series
4	Facility historical mean	Mean of all valid values for this indicator at this facility

This approach prevents temporary reporting gaps from creating artificial declines in service volumes.

Deviance Due to Incompleteness

Percent change in volume due to completeness adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	2.2%	4.7%	3.1%	4.7%	3.8%	2.2%	7.2%	9.0%	12.3%	18.4%	3.6%	8.3%
Bombali District	2.6%	3.3%	4.1%	10.4%	12.9%	7.4%	5.9%	5.6%	13.0%	5.6%	10.8%	8.6%
Bonthe District	1.5%	5.2%	3.6%	8.3%	4.3%	1.6%	4.1%	4.0%	24.3%	8.3%	2.7%	6.1%
Falaba District	0.5%	2.8%	3.0%	6.4%	4.2%	0.9%	0.3%	0.3%	12.8%	7.5%	1.8%	4.7%
Kailahun District	0.5%	1.9%	1.3%	13.8%	11.9%	0.5%	0.4%	0.4%	8.3%	8.1%	2.8%	4.4%
Kambia District	0.8%	2.8%	0.4%	1.4%	1.3%	0.5%	0.6%	0.8%	9.6%	5.3%	0.3%	1.4%
Karene District	0.3%	0.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	8.3%	6.3%	2.3%	4.8%
Kenema District	0.3%	1.2%	0.7%	1.7%	1.3%	0.1%	0.2%	0.2%	5.6%	4.5%	0.5%	1.1%
Koinadugu District	2.3%	3.9%	2.7%	4.3%	2.6%	1.5%	2.1%	2.3%	41.1%	23.9%	13.1%	14.9%
Kono District	0.4%	2.1%	1.2%	5.6%	3.1%	0.6%	0.7%	0.9%	6.8%	7.3%	1.2%	6.5%
Moyamba District	1.4%	3.8%	2.9%	4.0%	3.3%	1.7%	2.0%	2.2%	13.9%	7.4%	2.8%	5.3%
Port Loko District	1.5%	2.0%	2.1%	2.9%	4.8%	1.6%	1.3%	1.4%	19.9%	9.8%	10.3%	8.8%
Pujehun District	0.3%	1.1%	2.1%	5.2%	4.0%	0.1%	0.2%	0.3%	7.0%	3.9%	0.3%	0.9%
Tonkolili District	1.2%	3.0%	4.6%	6.5%	2.7%	1.5%	1.8%	1.7%	9.6%	8.2%	2.5%	3.7%
Western Area Rural District	0.7%	0.8%	1.7%	1.7%	2.0%	0.6%	2.6%	2.4%	22.7%	10.1%	0.2%	0.6%
Western Area Urban District	6.2%	8.6%	5.7%	15.9%	13.5%	5.7%	6.6%	6.4%	19.9%	15.5%	4.5%	6.4%

3% or above
1% to 2%
Below 1%

Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. The deviance is the difference in volume after imputing incomplete data. High levels of deviance can affect the plausability of the data.

Interpretation: Positive values indicate that missing data was imputed, increasing total reported volume. Values

When both adjustments are applied, outliers are corrected first, then missing values are imputed using the cleaned data.

Deviance Due to Incompleteness and Outliers

Percent change in volume due to both outlier and completeness adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	2.2%	4.5%	1.5%	4.7%	3.8%	2.2%	7.2%	9.0%	12.3%	18.4%	3.4%	8.1%
Bombali District	1.3%	3.3%	4.1%	8.1%	12.9%	7.4%	5.9%	5.6%	13.0%	4.7%	10.8%	8.6%
Bonthe District	1.5%	5.2%	3.6%	8.3%	4.3%	1.6%	4.1%	4.0%	21.8%	8.3%	2.7%	6.1%
Falaba District	0.5%	2.8%	3.0%	6.4%	4.2%	0.9%	0.4%	0.3%	12.8%	5.3%	1.8%	4.7%
Kailahun District	0.5%	1.9%	0.7%	13.8%	11.9%	0.5%	0.4%	0.4%	3.8%	5.9%	2.6%	4.2%
Kambia District	0.8%	2.8%	0.4%	1.4%	1.3%	0.5%	0.6%	0.8%	8.8%	5.3%	0.3%	1.4%
Karene District	0.3%	0.8%	0.0%	0.0%	0.0%	0.0%	2.7%	1.0%	2.6%	1.6%	2.3%	4.8%
Kenema District	0.3%	1.2%	0.7%	1.7%	1.3%	1.1%	1.5%	1.2%	3.7%	4.1%	0.5%	1.1%
Koinadugu District	2.3%	3.9%	2.7%	4.3%	2.3%	1.5%	2.1%	2.3%	38.6%	22.1%	13.1%	14.9%
Kono District	0.3%	2.1%	1.2%	5.6%	3.1%	0.6%	0.7%	0.9%	3.6%	6.8%	0.9%	4.7%
Moyamba District	1.4%	3.8%	2.9%	4.0%	3.3%	1.7%	2.0%	2.2%	10.2%	7.0%	2.7%	5.2%
Port Loko District	1.5%	2.0%	2.1%	2.9%	4.8%	1.6%	1.3%	1.4%	19.1%	9.8%	10.3%	8.8%
Pujehun District	0.3%	1.1%	2.1%	5.2%	4.0%	0.1%	0.2%	0.3%	7.0%	3.9%	0.0%	0.6%
Tonkolili District	1.2%	2.7%	4.6%	6.5%	2.7%	0.9%	1.8%	1.7%	8.1%	6.7%	2.4%	3.6%
Western Area Rural District	0.7%	0.8%	0.9%	0.2%	0.6%	0.1%	1.6%	2.4%	22.7%	10.1%	0.2%	0.6%
Western Area Urban District	6.2%	8.6%	5.7%	15.9%	13.5%	5.7%	6.0%	5.5%	18.7%	15.0%	3.0%	6.1%

3% or above
1% to 2%
Below 1%

Service utilization analysis - Module 3

Detecting and quantifying changes in health service delivery over time

Objectives

1. Measure changes in service volume

Year-over-year comparison of service volumes identifies increases or decreases across regions and indicators.

2. Detect and quantify disruptions

Statistical comparison of observed volumes against expected levels—derived from historical trends and seasonal patterns—enables identification and quantification of service shortfalls or surpluses.

Year-over-year change

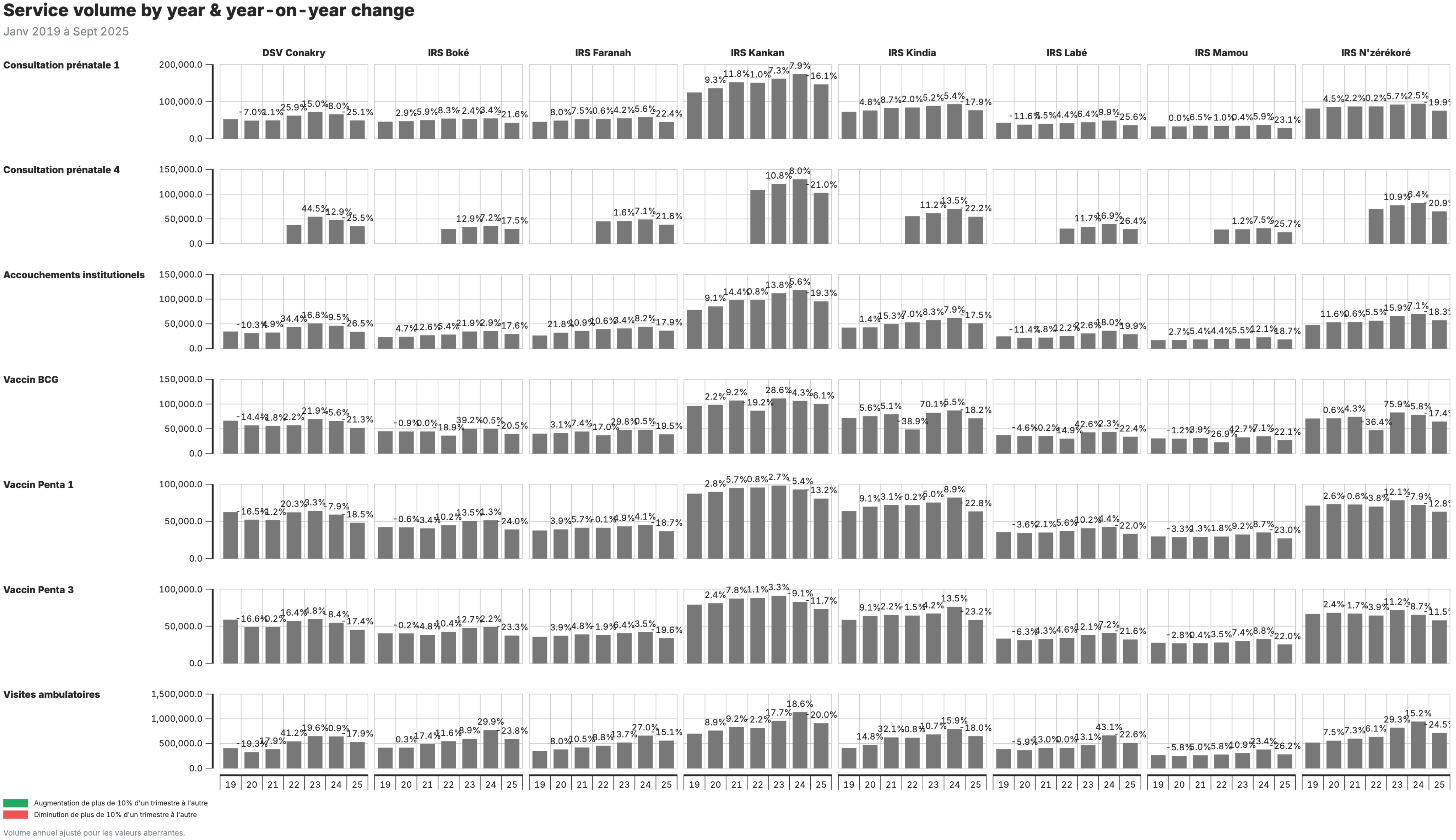
Year-over-year percent change quantifies shifts in service delivery between consecutive years.

For each indicator and region, total volume in the current year is compared to the previous year:

Percent change = (Current year − Previous year) ÷ Previous year × 100

Changes exceeding **±10%** are flagged for review.

Output: Change in service volume



Interpretation

Element	Description
Bars	Annual service volumes by region
Percentages	Year-over-year change annotations

Key considerations:

- Which regions exhibit the largest changes?
- Are changes consistent across regions or geographically concentrated?
- Do patterns vary by indicator?

Disruption detection

Our approach to service disruptions and surpluses utilizes an interrupted time series regression with facility-level fixed effects. Previous large and unexpected changes in historical data are removed. Unexpected volume changes are estimated by comparing observed volume to expected volume based on historical trends and seasonality.

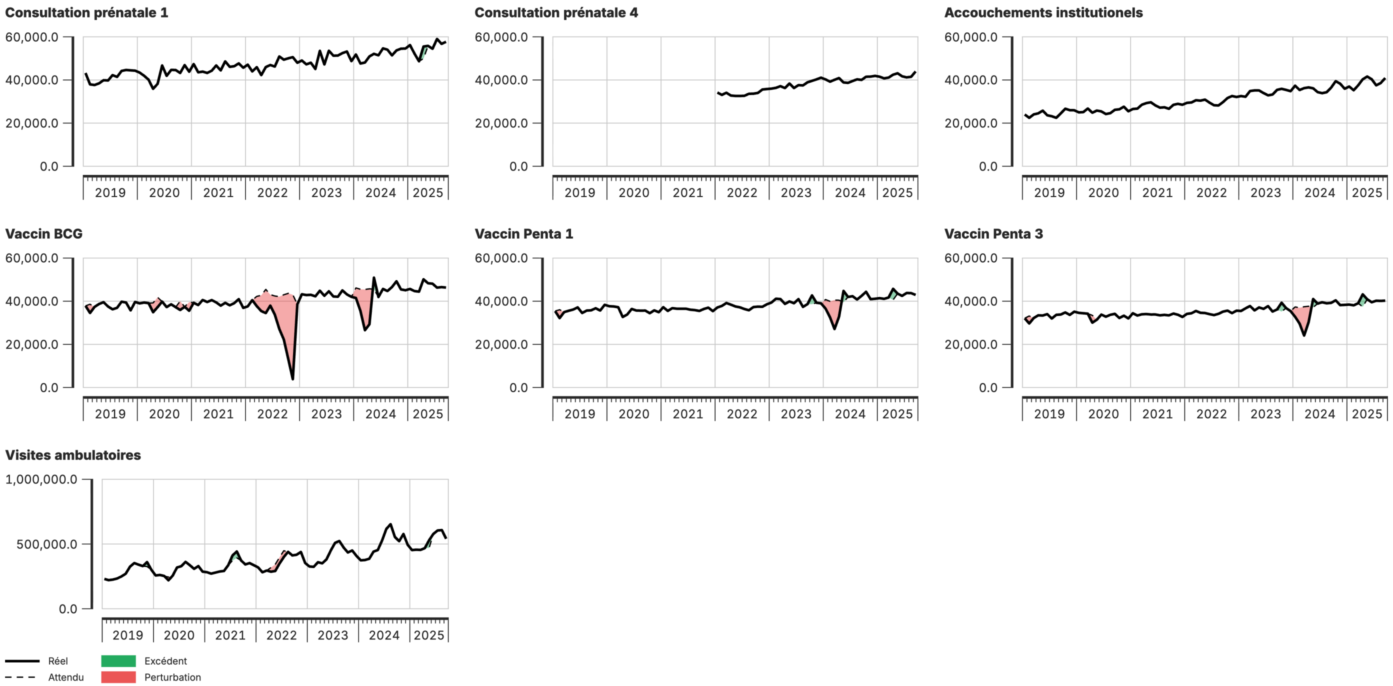
Disruptions and surpluses

Disruptions are flagged when volumes fall below anticipated levels, signaling potential barriers to access, resource shortages, or system failures.

Surpluses occur when volumes exceed expectations, which may indicate increased demand, over-reporting, or changes in service delivery.

Perturbations et excédents du volume de service, au niveau national

Janv 2019 à Sept 2025



Ce graphique quantifie les variations au niveau du volume de services, par rapport aux tendances historiques, en tenant compte de la saisonnalité. Ces signaux doivent être triangulés avec d'autres données et connaissances contextuelles, pour déterminer s'ils résultent d'une qualité de données inadéquate. Les variations inattendues de volume sont estimées en comparant le volume observé au volume attendu, selon les tendances historiques et la saisonnalité. Les variations inattendues importantes dans les données historiques sont exclues. Cette analyse repose sur une régression de séries chronologiques interrompues avec effets fixes au niveau des établissements.

How it works

Using past data to set expectations: We look at the past few years of data to understand the typical pattern for each month, accounting for regular seasonal changes.

Spotting unusual changes: We compare current service volumes to expectations. If we see volumes much higher or lower than expected, we flag it as an unusual change.

Handling past disruptions: We adjust historical data by removing previous big, unexpected changes so one-off events don't skew our understanding of what's "normal."

Detecting disruptions over time: We look at trends to see if there are clear shifts in health service use over several months.

Comparison to DHIS2

Extension of service utilization analysis, using more complex statistical approaches not available in DHIS2.

Using a regression framework, we are able to:

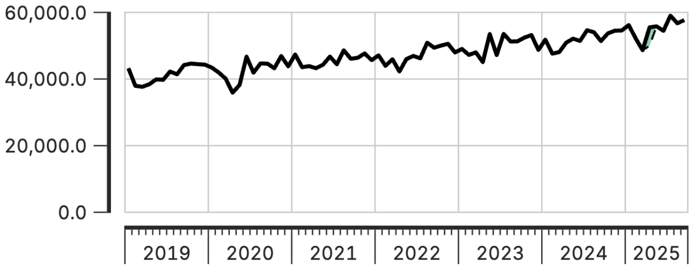
- Account for seasonality
- Exclude unusual changes to ensure one-off events aren't influencing normal trends
- Use historical data as a baseline for context
- Detect disruptions and recovery patterns
- Quantify changes with a robust methodology as compared to just observing simple fluctuations in a trend line

This improves the ability to interpret and compare utilization data across national and sub-national areas without needing population denominators.

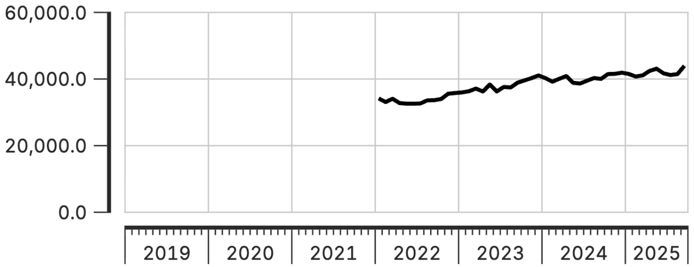
Perturbations et excédents du volume de service, au niveau national

Janv 2019 à Sept 2025

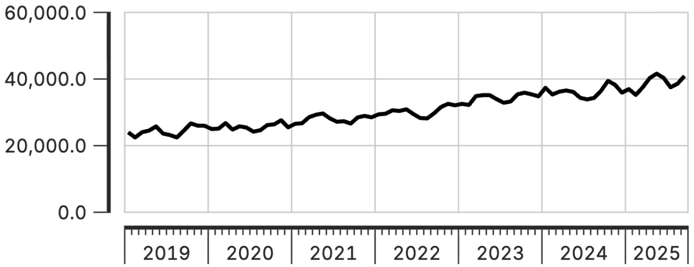
Consultation prénatale 1



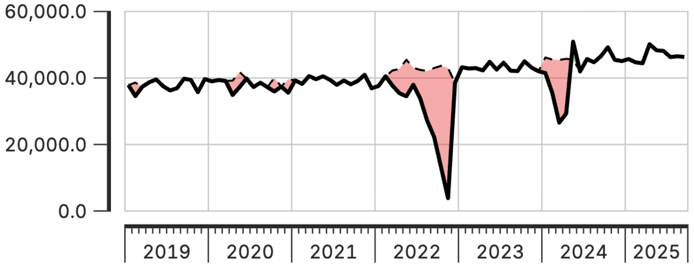
Consultation prénatale 4



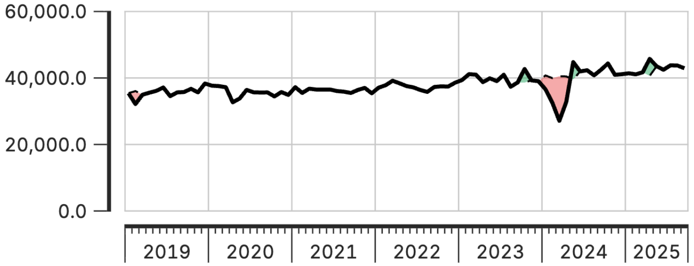
Accouchements institutionels



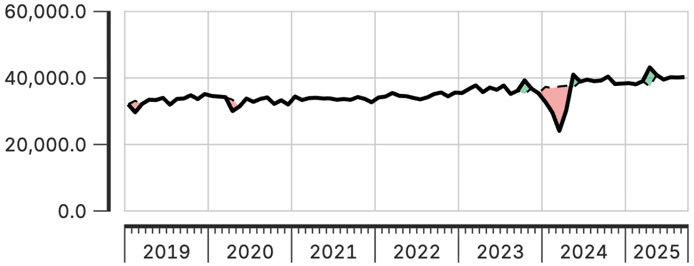
Vaccin BCG



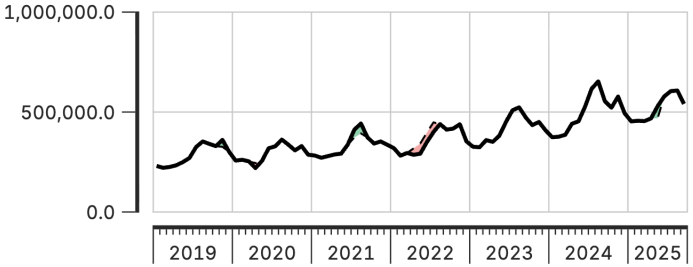
Vaccin Penta 1



Vaccin Penta 3



Visites ambulatoires

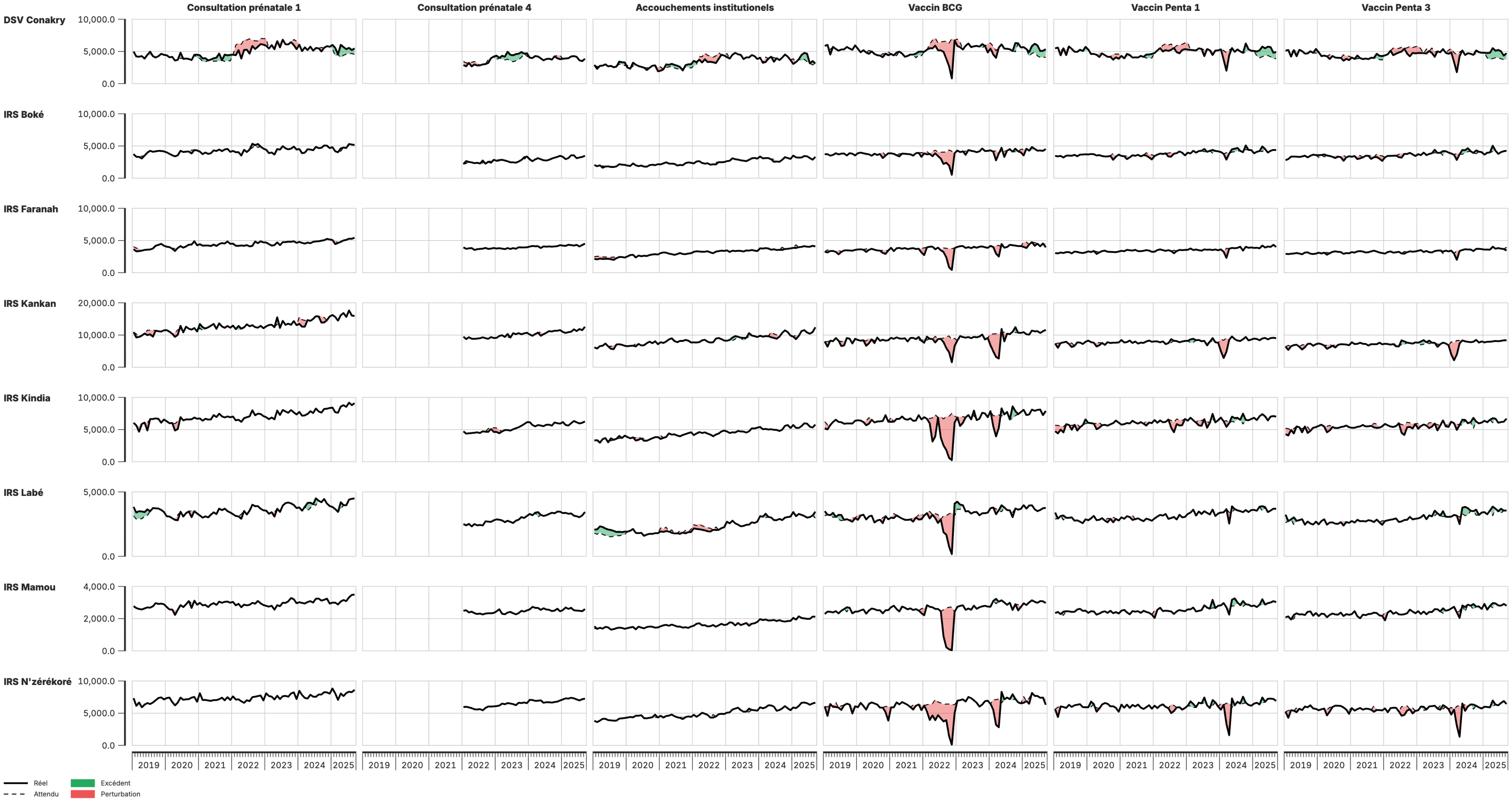


Réel
Attendu
Excédent
Perturbation

Ce graphique quantifie les variations au niveau du volume de services, par rapport aux tendances historiques, en tenant compte de la saisonnalité. Ces signaux doivent être triangulés avec d'autres données et connaissances contextuelles, pour déterminer s'ils résultent d'une qualité de données inadéquate. Les variations inattendues de volume sont estimées en comparant le volume observé au volume attendu, selon les tendances historiques et la saisonnalité. Les variations inattendues importantes dans les données historiques sont exclues. Cette analyse repose sur une régression de séries chronologiques interrompues avec effets fixes au niveau des établissements.

Perturbations et excédents du volume de service, au niveau infranational

Janv 2019 à Sept 2025



Ce graphique quantifie les variations au niveau du volume de services, par rapport aux tendances historiques, en tenant compte de la saisonnalité. Ces signaux doivent être triangulés avec d'autres données et connaissances contextuelles, pour déterminer s'ils résultent d'une qualité de données inadéquate. Les variations inattendues de volume sont estimées en comparant le volume observé au volume attendu, selon les tendances historiques et la saisonnalité. Les variations inattendues importantes dans les données historiques sont exclues. Cette analyse repose sur une régression de séries chronologiques interrompues avec effets fixes au niveau des établissements.

Service utilization module: Configuration parameters

Parameter	Description
Count variable to use for modeling	Which adjusted count to use for calculating expected values
Count variable to use for visualization	Which adjusted count to use as actual observed values
Run district-level model	Run regressions at admin_area_3 level. Set to Yes for detailed analysis, No for faster runtime
Run admin_area_4 analysis	Run finest-level analysis. Warning: can be very slow for large datasets
Threshold for MAD-based control limits	Number of MADs to flag sharp deviations. Default 1.5; higher = less sensitive
Smoothing window (k)	Window size in months for rolling median smoothing. Must be odd. Default 7
Dip threshold	Flag if actual falls below this proportion of expected. Default 0.9 (≥10% drop); use 0.8 to flag only big drops
Difference percent threshold	Highlight points where actual differs from expected by more than this percent. Default 10

Service coverage estimates - Module 4

Estimating the percentage of the target population that received a given health service

Our approach to service coverage analysis

Our approach derives and validates population denominators, significantly improving coverage estimates reported from HMIS systems.

In countries with accurate data, this approach helps identify subnational inequities and updates outdated estimates, while in countries with less precise data, the trends still provide valuable insights into performance.

We use these estimates to track recent trends and subnational disparities in the coverage of selected health services.

Two-part analytical process

The coverage estimation module operates in two sequential parts:

Part	Components
Part 1: Denominator calculation	Calculate target populations using multiple methods; compare against survey benchmarks; select optimal denominator for each indicator
Part 2: Coverage estimation	Apply denominator selections; project survey estimates forward using HMIS trends; generate final coverage estimates

What is service coverage?

Service coverage represents the proportion of the target population that received a specified health service.

$$\text{Service coverage} = \frac{\text{Population who received the service}}{\text{Population who need the service (target population)}}$$

Numerator comes from DHIS2 data, count of services adjusted for outliers

Several options for estimating the denominator

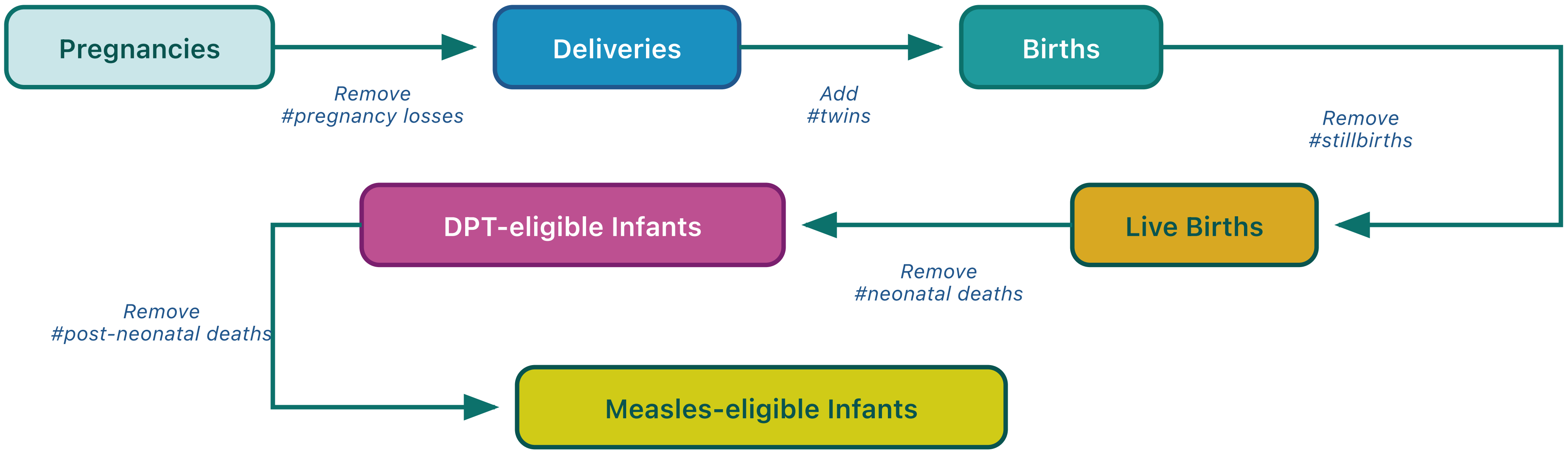
Denominators by service type

Each health indicator corresponds to a specific target population:

Service	Target population (denominator)
ANC1, ANC4	Pregnant women
Institutional delivery	Live births
BCG	Live births
Penta1, Penta3	Infants surviving beyond neonatal period
Measles1, Measles2	Infants surviving beyond infancy

Demographic cascade

Sequential demographic adjustments transform one target population estimate into another. Starting from pregnancies, demographic factors are applied to derive subsequent denominators:



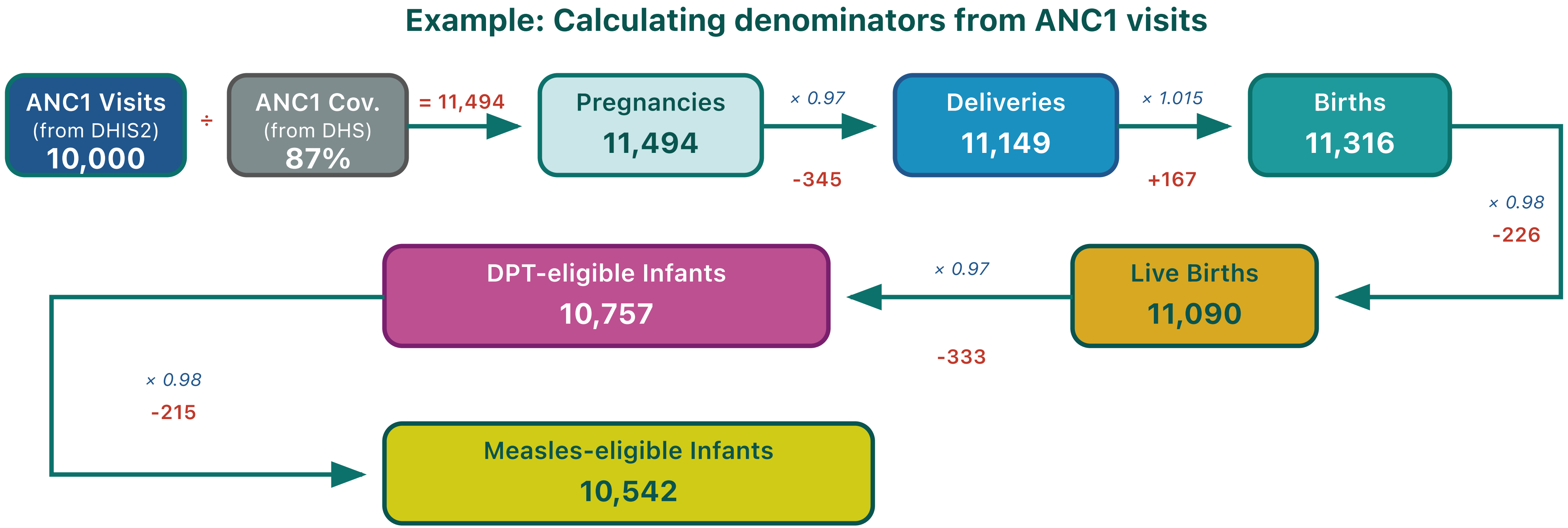
Default values:

Pregnancy loss rate = 0.03 | Twin rate = 0.015 | Stillbirth rate = 0.02 | NMR = 0.03 | PNMR = 0.02

Use country-specific values for stillbirth rate, NMR, PNMR, and IMR when available from DHS/MICS.

Denominator cascade: illustration

Starting from ANC1 service counts, demographic adjustment factors are applied sequentially to derive denominators for other services:



Calculation summary:

Step 1: Pregnancies = ANC1 visits ÷ ANC1 coverage = 10,000 ÷ 0.87 = 11,494

Step 2: Apply demographic factors through the cascade:

Pregnancies (11,494) → Deliveries (11,149) → Births (11,316) → Live Births (11,090) → DPT-eligible (10,757) → Measles-eligible (10,542)

Rates: Pregnancy loss = 3%, Twin rate = 1.5%, Stillbirth = 2%, NMR = 3%, PNMR = 2%

Forward and backward derivation

From any entry point, the cascade derives denominators in both directions:

Direction	Method	Example from Penta1
Forward	Apply mortality/attrition rates	DPT-eligible → Measles1-eligible → Measles2-eligible
Backward	Reverse mortality rates (add deaths back)	DPT-eligible → Live births → Births → Deliveries → Pregnancies

Backward derivation enables estimation of upstream populations from downstream service counts.

Denominators by entry point

Each HMIS indicator serves as an entry point. The module derives all target populations via forward and backward cascades:

Entry point	Base calculation	Forward derivation	Backward derivation
ANC1	ANC1 ÷ coverage → Pregnancies	Deliveries → Live births → DPT-eligible → Measles-eligible	—
Deliveries	Deliveries ÷ coverage → Deliveries	Live births → DPT-eligible → Measles-eligible	Pregnancies
BCG	BCG ÷ coverage → Live births	DPT-eligible → Measles-eligible	Deliveries → Pregnancies
Penta1	Penta1 ÷ coverage → DPT-eligible	Measles1-eligible → Measles2-eligible	Live births → Births → Deliveries → Pregnancies

Automatic denominator selection

For each indicator, the module selects the denominator that produces coverage closest to the survey benchmark.

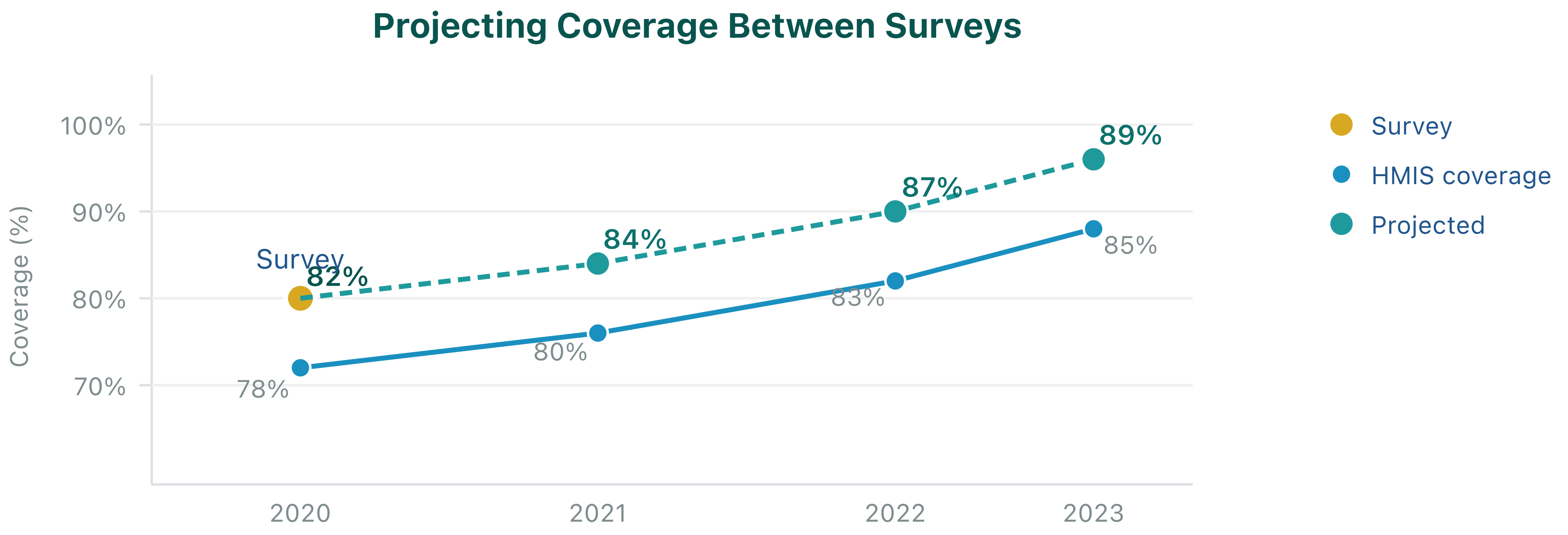
Selection algorithm:

1. Calculate coverage using each denominator option
2. Calculate squared error against survey: $(coverage - survey)^2$
3. Apply selection hierarchy (HMIS-based denominators prioritized over UN WPP)
4. Select the HMIS-based denominator with minimum error

Selection is made per indicator and geographic area. Users may override automatic selections in Part 2.

Coverage projection methodology

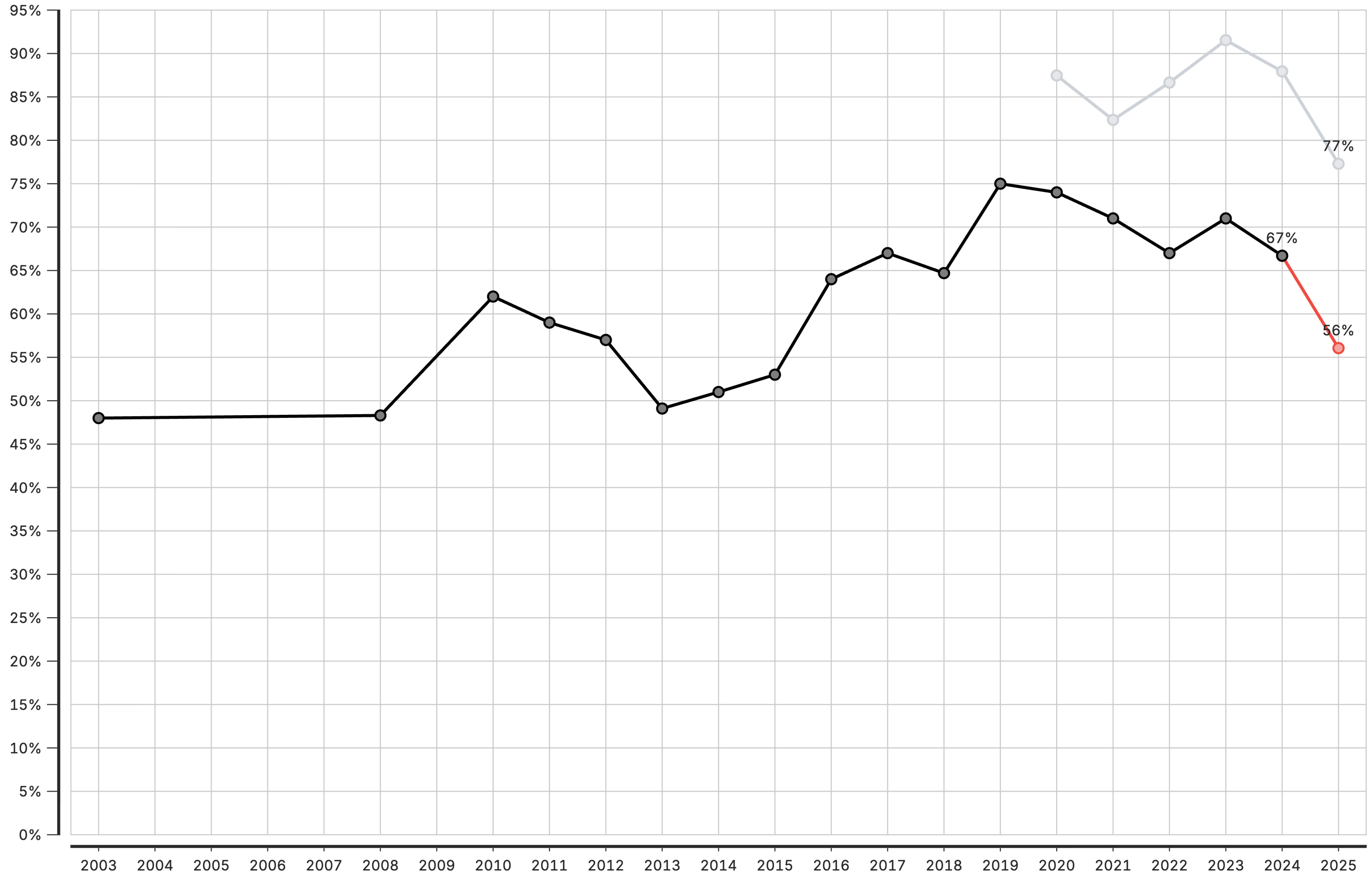
The module projects the most recent survey value forward using trends observed in HMIS-derived coverage:



Year-over-year changes (deltas) in HMIS coverage are calculated and applied to the last survey value. This approach preserves the survey baseline while incorporating observed service delivery trends.

Interpretation of coverage outputs

Element	Description
Black line/points	Survey data (DHS/MICS) — household survey reference
Grey line/points	HMIS-based coverage from facility data
Red line/points	Projected coverage — survey estimates extended using HMIS trends


95%
90%
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75%
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30%
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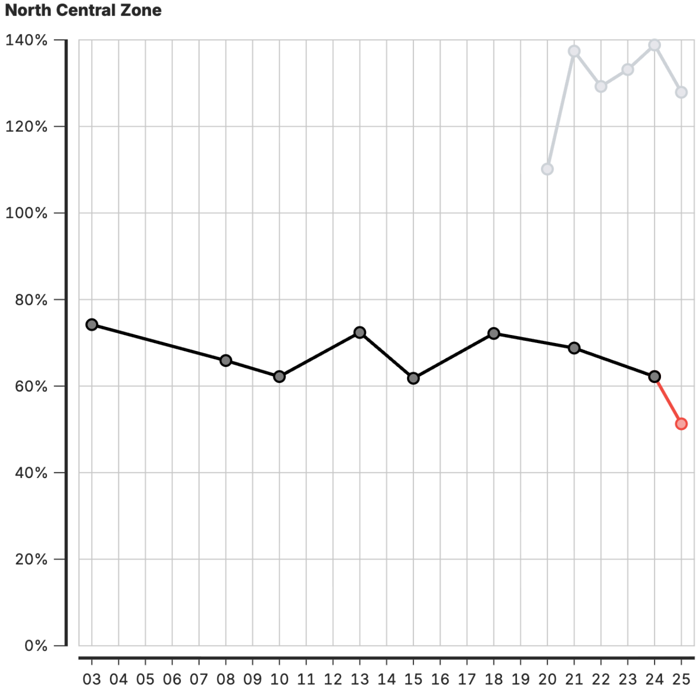
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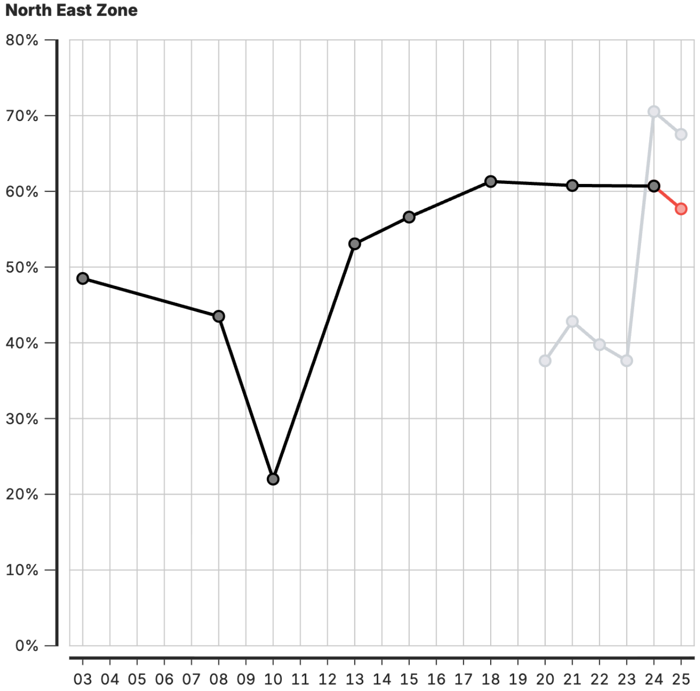
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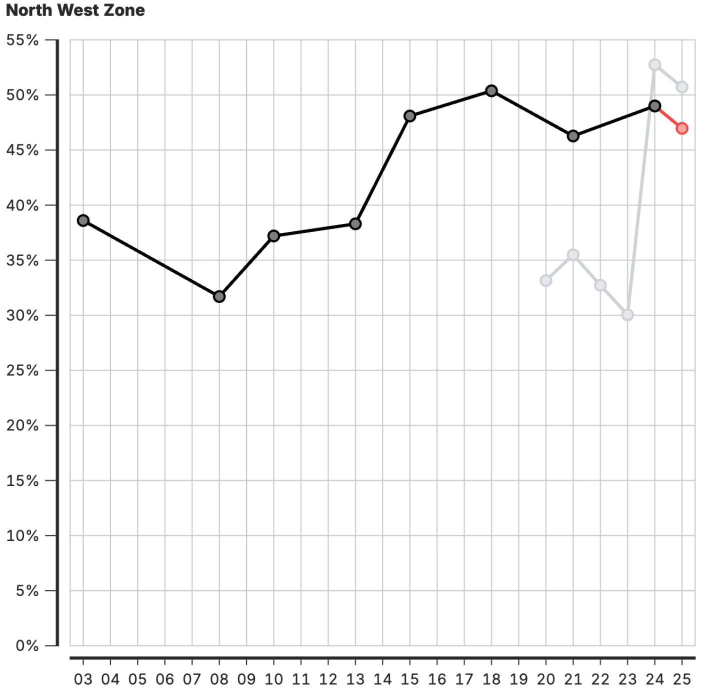
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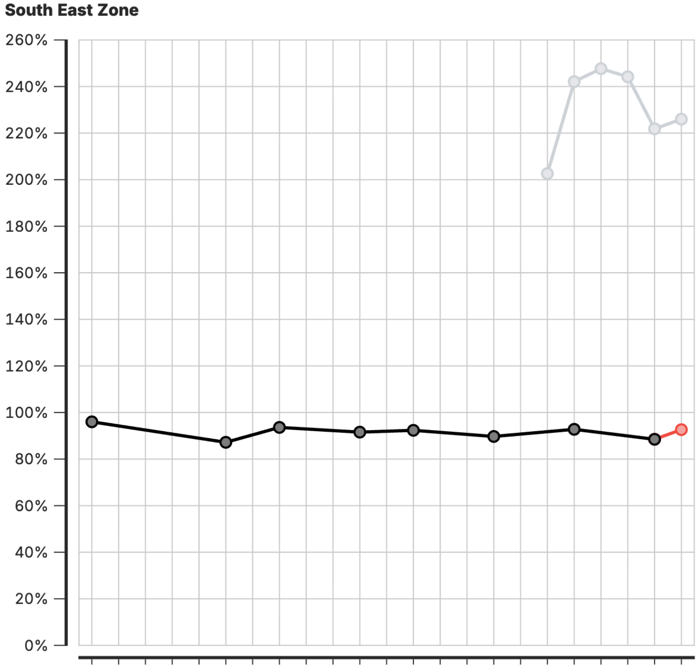
South East Zone
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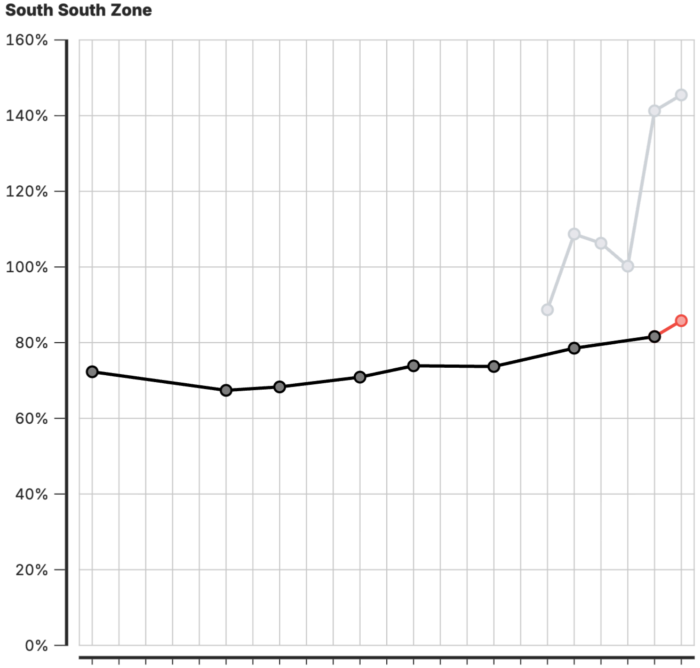
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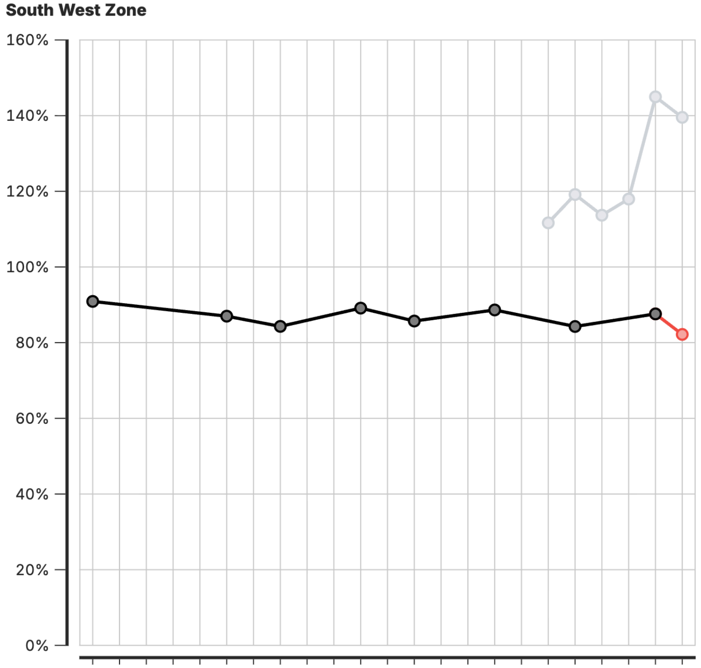
South West Zone
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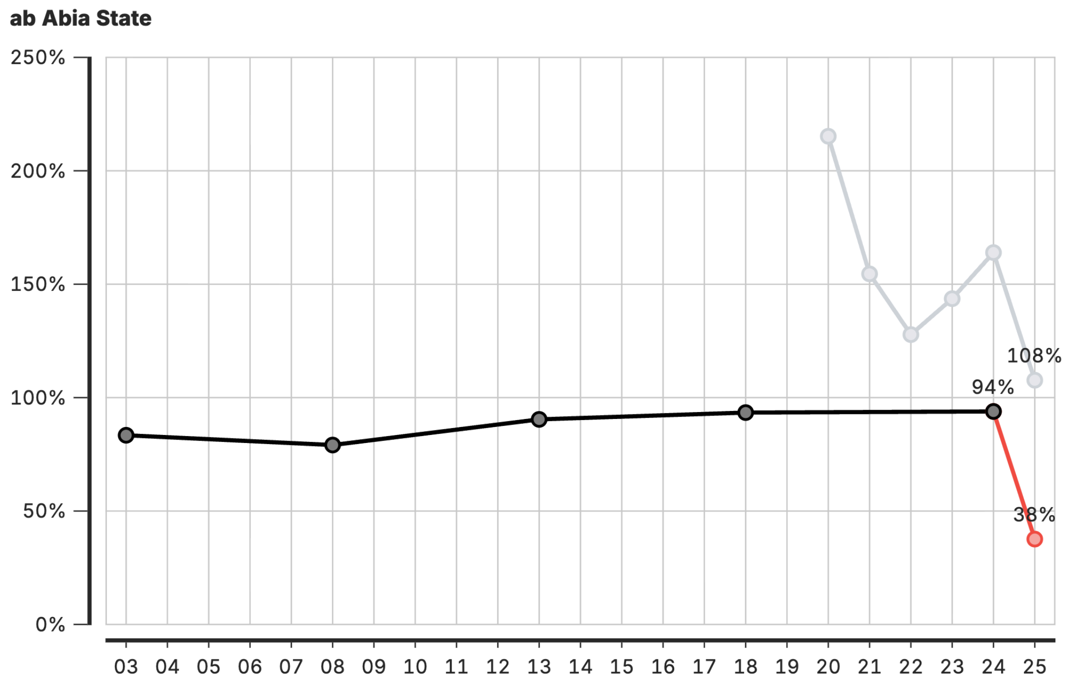
ab Abia State
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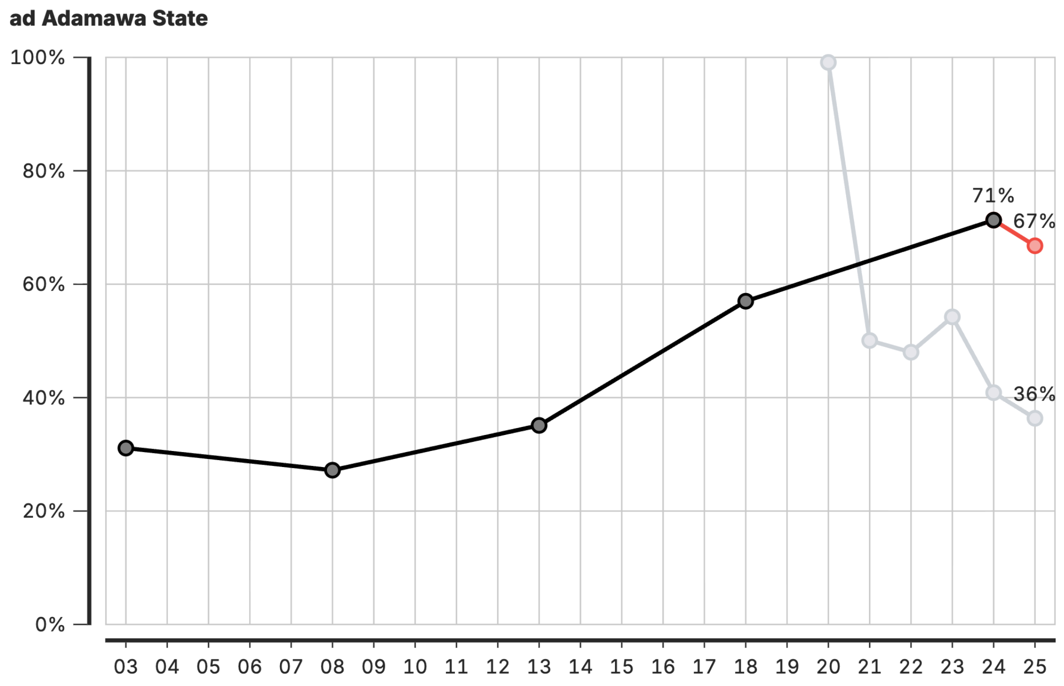
ad Adamawa State
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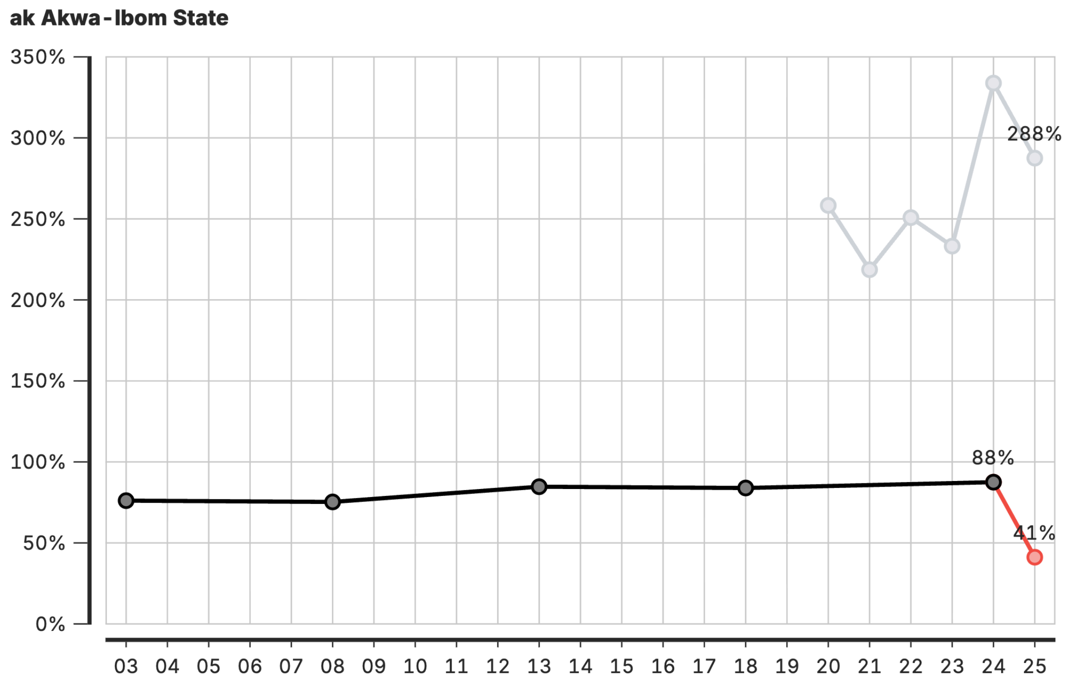
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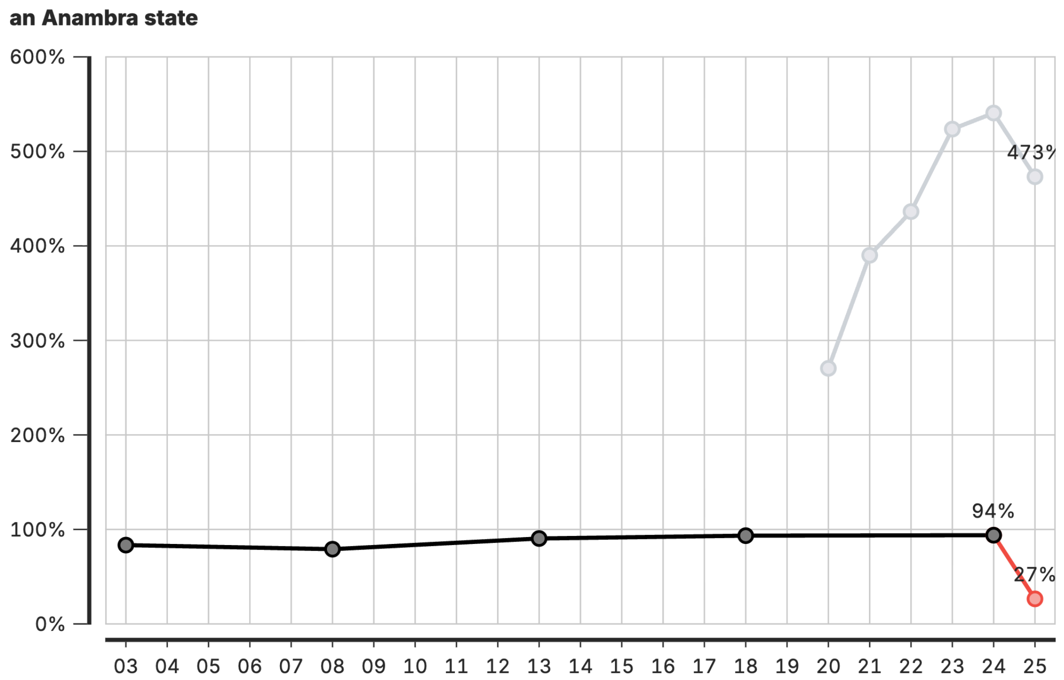
an Anambra state
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200%
100%
0%
94%
27%
03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Coverage module: Configuration parameters

Parameter	Description
Count value to use	Which adjusted count to use for coverage calculation
Level to calculate coverage for	Geographic levels for coverage estimation: national, provincial (admin area 2), or district (admin area 3)
Pregnancy loss rate	Proportion of pregnancies ending in loss before delivery
Twin rate	Proportion of deliveries resulting in twins
Stillbirth rate	Proportion of births that are stillbirths
Neonatal mortality rate	Deaths in first 28 days per live birth
Postneonatal mortality rate	Deaths from 28 days to 1 year per live birth
Infant mortality rate	Deaths before age 1 per live birth
Under 5 mortality rate	Deaths before age 5 per live birth

Country-specific mortality rates may be obtained from DHS reports, UN IGME, or national vital statistics.

Session 6: Creating a project

Activity: Creating a Project

In this hands-on session, we will:

- Set up a new project
- Configure project settings
- Select indicators and time periods
- Apply best practices for project organization

Participants will create their first project

Session 7: Creating visualizations

Activity: Creating Visualizations

In this hands-on session, we will:

- Explore available chart types
- Create and customize visualizations
- Export charts for use in reports

Participants will build visualizations from their analysis

🍽️ Lunch Break

90 minutes

Back at 14:00

Session 8: Creating reports

Activity: Creating Reports

In this hands-on session, we will:

- Use report templates
- Generate automated reports
- Customize report content and layout

Participants will create their first quarterly report draft

Day 3

Recap

On Day 2, we covered:

- FASTR methods for data quality assessment and adjustment
- Creating and configuring projects in the platform
- Building visualizations from Zambia data
- Generating reports

Day 3 Focus

Today we move from analysis to action:

- Deep dive into interpretation of results
- Create the Q4 2025 quarterly report
- Present findings to the group
- Develop action plans for continued work

Session 8: Interpretation of visualizations

Analytical thinking & interpretation

Content to be developed

This section will cover:

- Frameworks for interpreting FASTR outputs
- Connecting data patterns to programmatic meaning
- Common interpretation pitfalls to avoid
- Building analytical thinking skills

Session 9: Creating a Q4 2025 report

Generating quarterly reporting products

Content to be developed

This section will cover:

- Quarterly reporting workflow
- Using the FASTR platform for automated reports
- Quality assurance for reports
- Distribution and feedback mechanisms

🍽️ Lunch Break

90 minutes

Back at 14:00

Session 10: Presenting reports

End user mapping

End user mapping helps ensure that our outputs will meet the real needs of our end users.

Key questions

1. Who is my end user?
2. What does this end user need to accomplish with the report?
3. What information are they most interested in?
4. What do they like/not like about current reports?
5. How do they like to receive their information?

Day 4

Recap

On Day 3, we covered:

- Interpretation of FASTR visualizations
- Creation of the Q4 2025 quarterly report
- Presentation of findings and group feedback
- Action planning for continued platform use

Day 4 Focus

Today we design the Health Facility Assessment:

- Overview of the FASTR HFA phone survey
- Review questionnaire structure
- Adapt questionnaire for Zambia context
- Plan HFA priorities and data use

Session 12: Overview of FASTR HFA phone survey

Health Facility Assessment Phone Survey

Placeholder: Content to be developed

🍽️ Lunch Break

90 minutes

Back at 14:00

Next Steps & Action Planning

Key actions to take after this working session:

- Finalize adapted phone survey questionnaire
- Complete first quarterly report draft
- Train remaining MoH team on data downloader tools
- Prepare roadmap for participation in Abuja workshop

Quarterly Reporting Cycle

Establish a sustainable reporting rhythm:

- Define quarterly report timeline and responsibilities
- Set up data extraction schedule
- Establish review and approval process
- Plan dissemination to stakeholders

Continued Platform Usage

Maintain momentum with the FASTR platform:

- Regular data updates and quality checks
- Expanding analysis to additional indicators
- Building capacity across MoH teams
- Documentation of lessons learned

Workshop Closing

Thank you for your participation!

Over the past four days, we have:

- Configured the FASTR platform for Zambia
- Created the first quarterly RMNCAH-N report
- Adapted the HFA questionnaire for local context
- Built capacity for ongoing analysis and reporting

Stay Connected

Continue the journey:

- Platform access and support
- Quarterly report submissions
- Multi-country workshop in Abuja
- Ongoing technical assistance from GFF team

Thank You

FASTR Working Session - Zambia
January 27-30, 2026
Lusaka

Contact Information

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